

GRADE 5 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 6) • Forward 2 (Grade 7)
57 question sets • 1580 questions

CONTENTS

Set 1	5.OA.A.1	Evaluate expressions that use parentheses	6
Set 2	5.OA.A.1	Evaluate expressions with brackets and braces	43
Set 3	5.OA.A.1	Place grouping symbols to make an equation true	80
Set 4	5.OA.A.2	Write numerical expressions from words	117
Set 5	5.OA.A.2	Interpret numerical expressions without evaluating them	154
Set 6	5.OA.B.3	Generate two numerical patterns and compare their terms	191
Set 7	5.OA.B.3	Form ordered pairs from two patterns and graph them	222
Set 8	5.NBT.A.1	A digit is worth 10 times as much as the same digit one place to its right	259
Set 9	5.NBT.A.1	A digit is worth 1/10 of the same digit one place to its left	296
Set 10	5.NBT.A.2	Multiply and divide whole numbers by powers of 10; write powers of 10 with expon	333
Set 11	5.NBT.A.2	Place the decimal point when multiplying or dividing a decimal by a power of 10	364
Set 12	5.NBT.A.3.a	Read and write decimals to thousandths using numerals and words	395
Set 13	5.NBT.A.3.a	Write decimals to thousandths in expanded form	432
Set 14	5.NBT.A.3.b	Compare decimals to thousandths	463
Set 15	5.NBT.A.4	Round decimals to any place	494
Set 16	5.NBT.B.5	Multiply multi-digit whole numbers using the standard algorithm	525
Set 17	5.NBT.B.6	Divide with up to four-digit dividends and two-digit divisors	562

CONTENTS (continued)

Set 18	5.NBT.B.6	Explain division using area models and equations	599
Set 19	5.NBT.B.7	Add and subtract decimals to hundredths	636
Set 20	5.NBT.B.7	Multiply decimals to hundredths	673
Set 21	5.NBT.B.7	Divide decimals to hundredths	710
Set 22	5.NBT.B.7	Solve real-world problems with decimal operations	747
Set 23	5.NF.A.1	Add fractions with unlike denominators	778
Set 24	5.NF.A.1	Subtract fractions with unlike denominators	815
Set 25	5.NF.A.1	Add and subtract mixed numbers with unlike denominators	852
Set 26	5.NF.A.2	Solve word problems by adding and subtracting fractions	889
Set 27	5.NF.A.2	Use benchmark fractions to estimate and to check answers	926
Set 28	5.NF.B.3	Interpret a fraction as division of the numerator by the denominator	957
Set 29	5.NF.B.3	Solve sharing problems whose answers are fractions or mixed numbers	994
Set 30	5.NF.B.4.a	Multiply a fraction by a whole number	1025
Set 31	5.NF.B.4.a	Multiply a fraction by a fraction	1062
Set 32	5.NF.B.4.b	Find the area of a rectangle with fractional side lengths	1093
Set 33	5.NF.B.5.a	Compare the size of a product to the size of one factor without multiplying	1130
Set 34	5.NF.B.5.b	Explain the effect of multiplying by a fraction greater than, less than, or equal to 1	1161

CONTENTS (continued)

Set 35	5.NF.B.6	Solve real-world problems by multiplying fractions and mixed numbers	1192
Set 36	5.NF.B.6	Multiply mixed numbers	1229
Set 37	5.NF.B.7.a	Divide a unit fraction by a whole number	1266
Set 38	5.NF.B.7.b	Divide a whole number by a unit fraction	1303
Set 39	5.NF.B.7.c	Solve word problems that divide unit fractions and whole numbers	1334
Set 40	5.MD.A.1	Convert from a larger unit to a smaller unit	1365
Set 41	5.MD.A.1	Convert from a smaller unit to a larger unit	1402
Set 42	5.MD.A.1	Solve multistep problems that involve converting units	1439
Set 43	5.MD.B.2	Make a line plot of measurements in fractions of a unit	1470
Set 44	5.MD.B.2	Use operations on fractions to solve problems with line plot data	1507
Set 45	5.MD.C.3.a	Understand a unit cube and a cubic unit of volume	1544
Set 46	5.MD.C.3.b	A solid packed with n unit cubes has a volume of n cubic units	1575
Set 47	5.MD.C.4	Measure volume by counting unit cubes	1606
Set 48	5.MD.C.5.a	Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$	1637
Set 49	5.MD.C.5.b	Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems	1668
Set 50	5.MD.C.5.b	Find a missing dimension when the volume is known	1699
Set 51	5.MD.C.5.c	Find the volume of a figure made of two rectangular prisms	1730

CONTENTS (continued)

Set 52	5.G.A.1	Understand the coordinate plane: axes, origin, and coordinates	1761
Set 53	5.G.A.2	Graph points in the first quadrant	1792
Set 54	5.G.A.2	Solve real-world problems by graphing points and interpreting coordinates	1823
Set 55	5.G.B.3	Attributes of a category belong to all of its subcategories	1854
Set 56	5.G.B.4	Classify quadrilaterals in a hierarchy based on their properties	1891
Set 57	5.G.B.4	Classify triangles by their sides and angles	1922

Evaluate expressions that use parentheses

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting multi-digit numbers	4.NBT.B.4
FORWARD 1	Evaluate expressions with variables	6.EE.A.2.c
FORWARD 2	Evaluate expressions with negative numbers	7.NS.A.3

SET 1 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (8 + 5)$$

Value: _____

Evaluate.

$$(35 - 11) \div 4$$

Value: _____

Evaluate.

$$50 - (6 \times 7)$$

Value: _____

Evaluate.

$$3 + (24 \div 6) \times 5$$

A. 35

B. 135

C. 27

D. 23

Evaluate.

$$(9 + 3) \times (10 - 4)$$

Value: _____

SET 1 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

In $4 \times (2 + 5)$, you add $2 + 5$ first.

A. True

B. False

Evaluate.

$$(6 + 4) \times 2$$

Value: _____

Which step comes first in $18 - (3 \times 4)$?

- A.** Subtract $18 - 4$
- B.** Multiply 3×4
- C.** Subtract $18 - 3$
- D.** Add $3 + 4$

Evaluate.

$$20 - (5 + 5)$$

Value: _____

True or false?

$$(10 - 4) + 3 = 10 - 7$$

A. True

B. False

SET 1 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 8$$

Product: _____

Divide.

$$24 \div 6$$

Quotient: _____

True or false?

$$9 \times 7 = 56$$

A. True

B. False

$$48 \div 8 = ?$$

A. 8

B. 5

C. 7

D. 6

Multiply.

$$3 \times 9$$

Product: _____

SET 1 • 5.OA.A.1

BACKWARD 3

Adding and subtracting multi-digit numbers • Grade 4 • Number & Operations
in Base Ten • 4.NBT.B.4

Add.

$$348 + 275$$

Sum: _____

Subtract.

$$502 - 168$$

Difference: _____

True or false?

$$1,250 - 375 = 985$$

A. True

B. False

$$609 + 194 = ?$$

A. 793

B. 703

C. 813

D. 803

Subtract.

$$1,000 - 432$$

Difference: _____

SET 1 • 5.OA.A.1

FORWARD 1

Evaluate expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4(x + 3)$ when $x = 5$.

Value: _____

Evaluate $20 - 2y$ when $y = 6$.

Value: _____

Evaluate $(a + b) \div 3$ when $a = 10$ and $b = 8$.

- A.** 4
- B.** 9
- C.** 54
- D.** 6

Evaluate $3m + 2$ when $m = 7$.

Value: _____

True or false?

When $n = 4$, the value of $5(n - 1)$ is 19.

A. True

B. False

SET 1 • 5.OA.A.1

FORWARD 2

Evaluate expressions with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$-3 \times (4 - 9)$$

Value: _____

Find the value.

$$(-12 + 4) \div 2$$

Value: _____

Find the value.

$$6 - (2 - 10)$$

A. -2

B. 2

C. 14

D. 18

Find the value.

$$(-5) \times (-2) + (-7)$$

Value: _____

True or false?

$$(-20) \div (3 + 2) = 4$$

A. True

B. False

Evaluate expressions with brackets and braces

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting within 100	2.NBT.B.5
FORWARD 1	Evaluate expressions with whole-number exponents	6.EE.A.1
FORWARD 2	Grouping symbols with negative numbers	7.NS.A.3

SET 2 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times [5 + (3 \times 4)]$$

Value: _____

Evaluate.

$$[20 - (2 + 6)] \div 3$$

Value: _____

Evaluate.

$$4 \times \{3 + [10 - (2 \times 4)]\}$$

A. 36

B. 20

C. 44

D. 26

Evaluate.

$$60 \div [3 \times (7 - 5)]$$

Value: _____

Evaluate.

$$\{18 - [2 \times (4 + 1)]\} + 9$$

Value: _____

SET 2 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(12 - 5) \times 3$$

Value: _____

True or false?

In $(8 + 2) \div 5$, you divide $2 \div 5$ first.

A. True

B. False

Evaluate.

$$4 \times (9 - 6)$$

A. 33

B. 10

C. 30

D. 12

Evaluate.

$$30 \div (2 + 4)$$

Value: _____

True or false?

$$(3 + 5) \times 2 = 16$$

A. True

B. False

SET 2 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$8 \times 6$$

Product: _____

Divide.

$$56 \div 7$$

Quotient: _____

True or false?

$$36 \div 4 = 8$$

A. True

B. False

$$7 \times 9 = ?$$

A. 63

B. 56

C. 72

D. 49

Divide.

$$81 \div 9$$

Quotient: _____

SET 2 • 5.OA.A.1

BACKWARD 3

Adding and subtracting within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$46 + 37$$

Sum: _____

Subtract.

$$72 - 28$$

Difference: _____

True or false?

$$55 + 29 = 74$$

A. True

B. False

$$90 - 47 = ?$$

A. 47

B. 43

C. 53

D. 33

Add.

$$38 + 54$$

Sum: _____

SET 2 • 5.OA.A.1

FORWARD 1

Evaluate expressions with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2 \times (3 + 1)^2$$

Value: _____

Evaluate.

$$[10 - 2^3] \times 5$$

Value: _____

Evaluate.

$$4^2 - (6 + 2)$$

A. 8

B. 2

C. 0

D. 24

Evaluate.

$$(5^2 - 1) \div 6$$

Value: _____

True or false?

$$3 \times 2^2 = 36$$

A. True

B. False

SET 2 • 5.OA.A.1

FORWARD 2

Grouping symbols with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$2 \times [-3 + (8 \div (-4))]$$

Value: _____

Find the value.

$$[(-6) + 15] \div (-3)$$

Value: _____

Find the value.

$$-4 \times [5 - (-1)]$$

A. 16

B. -16

C. -24

D. 24

Find the value.

$$\{10 - [2 \times (-3)]\} \div 4$$

Value: _____

True or false?

$$[-8 - (-2)] \times 3 = -30$$

A. True

B. False

Place grouping symbols to make an equation true

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	When regrouping keeps the same value	3.OA.B.5
BACKWARD 3	Multiplication and division facts	3.OA.C.7
FORWARD 1	Use parentheses to write equivalent expressions	6.EE.A.3
FORWARD 2	Expand and factor expressions with rational coefficients	7.EE.A.1

SET 3 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Insert parentheses to make the equation true.

$$4 + 6 \times 3 = 30$$

Equation: _____

Insert parentheses to make the equation true.

$$20 - 8 - 2 = 14$$

Equation: _____

Which placement of parentheses makes the equation true?

$$2 \times 5 + 3 \times 4 = 52$$

A. $2 \times (5 + 3) \times 4 = 52$

B. $(2 \times 5 + 3) \times 4 = 52$

C. $2 \times (5 + 3 \times 4) = 52$

D. $2 \times 5 + (3 \times 4) = 52$

Insert parentheses to make the equation true.

$$36 \div 6 - 2 = 9$$

Equation: _____

Insert brackets and parentheses to make the equation true.

$$2 \times 9 - 3 + 1 = 10$$

Equation: _____

SET 3 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(4 + 6) \times 3$$

Value: _____

Evaluate.

$$4 + (6 \times 3)$$

Value: _____

True or false?

$$(20 - 8) - 2 = 20 - (8 - 2)$$

A. True

B. False

Evaluate.

$$36 \div (6 - 2)$$

A. 4

B. 34

C. 9

D. 7

Evaluate.

$$(36 \div 6) - 2$$

Value: _____

SET 3 • 5.OA.A.1

BACKWARD 2

When regrouping keeps the same value • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

True or false?

$$(12 \div 6) \div 2 = 12 \div (6 \div 2)$$

A. True

B. False

Which equation is true?

A. $(5 + 2) + 3 = 5 + (2 + 3)$

B. $(9 - 4) - 1 = 9 - (4 - 1)$

C. $(16 \div 4) \div 2 = 16 \div (4 \div 2)$

D. $(7 - 3) - 2 = 7 - (3 - 2)$

True or false?

$$4 \times (5 \times 2) = (4 \times 5) \times 2$$

A. True

B. False

Fill in the blank.

$$(6 \times 2) \times 5 = 6 \times (2 \times \underline{\quad})$$

Blank: _____

SET 3 • 5.OA.A.1

BACKWARD 3

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 4$$

Product: _____

Divide.

$$42 \div 6$$

Quotient: _____

True or false?

$$8 \times 7 = 54$$

A. True

B. False

$$64 \div 8 = ?$$

A. 6

B. 7

C. 8

D. 9

Multiply.

$$6 \times 7$$

Product: _____

SET 3 • 5.OA.A.1

FORWARD 1

Use parentheses to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(x + 5)$$

Expression: _____

Expand.

$$4(2y - 1)$$

Expression: _____

Which expression is equivalent to $6(a + 2)$?

- A.** $6a + 2$
- B.** $a + 12$
- C.** $6a + 12$
- D.** $8a$

Factor.

$$5m + 20$$

Expression: _____

True or false?

$$2(x + 7) = 2x + 7$$

A. True

B. False

SET 3 • 5.OA.A.1

FORWARD 2

Expand and factor expressions with rational coefficients • Grade 7 •
Expressions & Equations • 7.EE.A.1

Expand.

$$-2(3x - 4)$$

Expression: _____

Expand.

$$\frac{1}{2}(6y + 10)$$

Expression: _____

Which expression is equivalent to $-5(n + 2)$?

A. $-5n + 2$

B. $-5n - 10$

C. $-5n + 10$

D. $5n - 10$

Factor out 3.

$$3x - 12$$

Expression: _____

True or false?

$$0.5(4a - 8) = 2a - 8$$

A. True

B. False

Write numerical expressions from words

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Quotients as equal shares	3.OA.A.2
BACKWARD 3	Sums and differences	2.OA.A.1
FORWARD 1	Write expressions with variables	6.EE.A.2.a
FORWARD 2	Write and solve equations from words	7.EE.B.4.a

SET 4 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 9 and 6, then multiply by 4."

Expression: _____

Write an expression for "subtract 5 from 30, then divide by 5."

Expression: _____

Which expression shows "triple the sum of 12 and 8"?

A. $12 + 8 \times 3$

B. $3 \times 12 + 8$

C. $3 \times (12 + 8)$

D. $(3 + 12) \times 8$

Write an expression for "divide 40 by 8, then add 7."

Expression: _____

Write an expression for "the product of 6 and the difference of 15 and 9."

Expression: _____

SET 4 • 5.OA.A.2

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

True or false?

5×7 can mean 5 groups of 7.

A. True

B. False

Which expression shows 4 groups of 6?

A. 46

B. $6 - 4$

C. $4 + 6$

D. 4×6

Write a multiplication expression for 8 groups of 3.

Expression: _____

True or false?

$$3 + 3 + 3 + 3 = 3 \times 3$$

A. True

B. False

What is the product of 7 and 5?

Product: _____

SET 4 • 5.OA.A.2

BACKWARD 2

Quotients as equal shares • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

True or false?

$15 \div 3$ can mean 15 shared equally into 3 groups.

A. True

B. False

Which expression shows 20 apples shared equally among 4 bags?

A. 20×4

B. $4 \div 20$

C. $20 \div 4$

D. $20 - 4$

What is the quotient of 18 and 6?

Quotient: _____

Write a division expression for 32 split into groups of 8.

Expression: _____

True or false?

The quotient of 12 and 2 is 24.

- A. True
- B. False

SET 4 • 5.OA.A.2

BACKWARD 3

Sums and differences • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

What is the sum of 14 and 9?

Sum: _____

What is the difference of 20 and 7?

Difference: _____

True or false?

The difference of 15 and 6 is 21.

A. True

B. False

Which word names the answer to an addition problem?

- A.** Difference
- B.** Quotient
- C.** Product
- D.** Sum

Write an expression for "7 more than 25."

Expression: _____

SET 4 • 5.OA.A.2

FORWARD 1

Write expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for "the sum of n and 8, times 3."

Expression: _____

Write an expression for "y divided by 5, then minus 2."

Expression: _____

Which expression means "4 less than the product of 2 and x"?

A. $4 - 2x$

B. $2x - 4$

C. $2(x - 4)$

D. $2 + x - 4$

Write an expression for "twice the difference of m and 6."

Expression: _____

True or false?

"9 more than k " can be written as $k + 9$.

A. True

B. False

SET 4 • 5.OA.A.2

FORWARD 2

Write and solve equations from words • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Three times the sum of a number and 4 is 27.

Write and solve an equation to find the number.

Equation and solution: _____

Half of a number, minus 6, is 10.

Write and solve an equation to find the number.

Equation and solution: _____

Which equation represents "2 less than 5 times a number is 33"?

A. $5n + 2 = 33$

B. $5n - 2 = 33$

C. $2 - 5n = 33$

D. $5(n - 2) = 33$

A number divided by 4, plus 3, is 9.

Write and solve an equation to find the number.

Equation and solution: _____

True or false?

The solution of $2(x + 3) = 18$ is $x = 7$.

A. True

B. False

Interpret numerical expressions without evaluating them

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Multiplicative comparison	4.OA.A.1
BACKWARD 2	Groups of a sum	3.OA.A.1
BACKWARD 3	Equal shares	3.OA.A.2
FORWARD 1	Identify parts of an expression	6.EE.A.2.b
FORWARD 2	Rewrite expressions to show how quantities are related	7.EE.A.2

SET 5 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

How does $4 \times (563 + 218)$ compare to $563 + 218$?

- A.** It is 4 more.
- B.** It is 4 times as large.
- C.** It is 4 less.
- D.** It is one-fourth as large.

True or false?

$(7,215 - 1,005) \div 5$ is one-fifth as large as $7,215 - 1,005$.

A. True

B. False

Without calculating, how many times as large is $9 \times (48 + 27)$ as $48 + 27$?

Times as large: _____

Which expression is twice as large as 356×12 ?

A. $(356 + 2) \times 12$

B. $356 \times 12 + 2$

C. 356×14

D. $2 \times (356 \times 12)$

True or false?

$6 \times 25 + 4$ is 6 times as large as $25 + 4$.

A. True

B. False

SET 5 • 5.OA.A.2

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.1

True or false?

35 is 5 times as many as 7.

A. True

B. False

Fill in the blank.

48 is ____ times as many as 6.

Blank: _____

Which equation shows "36 is 4 times as many as 9"?

A. $9 = 4 \times 36$

B. $36 = 4 + 9$

C. $36 - 9 = 4$

D. $36 = 4 \times 9$

True or false?

30 is 3 times as many as 27.

A. True

B. False

Tom has 6 cards. Ana has 3 times as many cards.
How many cards does Ana have?

Cards: _____

SET 5 • 5.OA.A.2

BACKWARD 2

Groups of a sum • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 1)$ means 3 groups of $(4 + 1)$.

A. True

B. False

Which expression means 6 groups of $(10 + 2)$?

A. $6 \times (10 + 2)$

B. $(6 + 10) \times 2$

C. $6 \times 10 + 2$

D. $6 + (10 + 2)$

How many groups of $(5 + 4)$ are in $7 \times (5 + 4)$?

Groups: _____

True or false?

$$2 \times (8 + 1) = (8 + 1) + 2$$

A. True

B. False

Write 4 groups of 9 as a multiplication expression.

Expression: _____

SET 5 • 5.OA.A.2

BACKWARD 3

Equal shares • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

True or false?

12 shared equally among 3 people is 4 each.

A. True

B. False

A 20-inch ribbon is cut into 5 equal pieces.

How long is each piece?

Length: _____

Which expression finds one of 4 equal shares of 28?

A. $28 - 4$

B. $28 \div 4$

C. 28×4

D. $4 \div 28$

True or false?

Dividing a number by 2 splits it into 2 equal shares.

A. True

B. False

30 crayons are shared equally among 6 children.

How many crayons does each child get?

Crayons: _____

SET 5 • 5.OA.A.2

FORWARD 1

Identify parts of an expression • Grade 6 • Expressions & Equations •
6.EE.A.2.b

In $5(x + 2)$, what is 5?

- A.** A term of $x + 2$
- B.** A quotient
- C.** A sum
- D.** A factor

How many terms are in $4a + 7b - 1$?

Terms: _____

What is the coefficient of m in $9m + 4$?

Coefficient: _____

True or false?

In $3(y + 6)$, the quantity $(y + 6)$ is a sum of two terms.

A. True

B. False

What is the expression $12 \div (n + 1)$ called?

- A.** A coefficient
- B.** A quotient
- C.** A sum
- D.** A product

SET 5 • 5.OA.A.2

FORWARD 2

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price p is increased by 5%.

Which expression shows the new price?

A. $0.05p$

B. $1.05p$

C. $5p$

D. $p + 5$

True or false?

$$x + 0.3x = 0.4x$$

A. True

B. False

Write $n - 0.4n$ as one term.

Expression: _____

Which expression is equivalent to $2(3t)$?

- A.** $5t$
- B.** $2t + 3t$
- C.** $6t$
- D.** $3t + 2$

A sale price is $0.75p$, where p is the original price.
What percent of the original price is taken off?

Percent: _____

Generate two numerical patterns and compare their terms

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Equations relating dependent and independent variables	6.EE.C.9
FORWARD 2	Equations for proportional relationships	7.RP.A.2.c

SET 6 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 3.

Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Pattern A: _____

Pattern B: _____

Pattern A: 0, 4, 8, 12, 16

Pattern B: 0, 12, 24, 36, 48

How does each term in Pattern B compare to the matching term in Pattern A?

A. 8 more

B. 12 more

C. 4 times as large

D. 3 times as large

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10.

What is the 6th term of each pattern?

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8. Each term in Pattern B is 4 times the matching term in Pattern A.

- A.** True
- B.** False

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

When Pattern A is 28, what is the matching term in Pattern B?

Term: _____

SET 6 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 6." The pattern starts at 0.
Write the first five terms.

Terms: _____

True or false?

The rule "add 4," starting at 0, gives 0, 4, 8, 16.

A. True

B. False

What is the rule for the pattern?

0, 9, 18, 27, 36

- A.** Add 9
- B.** Add 18
- C.** Multiply by 9
- D.** Add 8

The rule is "add 11." The pattern starts at 0.
What is the 4th term?

Term: _____

What comes next?

0, 15, 30, 45, ____

Next term: _____

SET 6 • 5.OA.B.3

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$4 \times 7$$

Product: _____

True or false?

$$6 \times 8 = 42$$

A. True

B. False

What number times 3 equals 24?

- A.** 8
- B.** 7
- C.** 21
- D.** 6

Multiply.

$$5 \times 9$$

Product: _____

What number times 6 equals 42?

Number: _____

SET 6 • 5.OA.B.3

FORWARD 1

Equations relating dependent and independent variables • Grade 6 •
Expressions & Equations • 6.EE.C.9

Each box holds 6 cans.

Write an equation for the number of cans c in b boxes.

Equation: _____

In $c = 6b$, which variable is the dependent variable?

- A.** b
- B.** c
- C.** 6
- D.** Neither

Write an equation for y in terms of x .

x	1	2	3	4
y	5	10	15	20

Equation: _____

True or false?

In $y = x + 4$, y is the independent variable.

A. True

B. False

A plant grows 2 cm each week.

Write an equation for its growth g after w weeks.

Equation: _____

SET 6 • 5.OA.B.3

FORWARD 2

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost \$0.35 each.

Write an equation for the cost c of n pencils.

Equation: _____

The table shows a proportional relationship.

Write an equation for y in terms of x .

x	4	6	10
y	10	15	25

Equation: _____

Which equation represents a proportional relationship?

A. $y = 1.5$

B. $y = x + 1.5$

C. $y = 1.5x + 2$

D. $y = 1.5x$

A train travels 150 miles in 2 hours at a constant speed.
Write an equation for the distance d after t hours.

Equation: _____

True or false?

$y = \frac{3}{4}x$ represents a proportional relationship.

A. True

B. False

Form ordered pairs from two patterns and graph them

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Ordered pairs and the coordinate plane	5.G.A.1
BACKWARD 3	Reading the scale on an axis	3.MD.B.3
FORWARD 1	Plot pairs of values from a ratio table	6.RP.A.3.a
FORWARD 2	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a

SET 7 • 5.OA.B.3

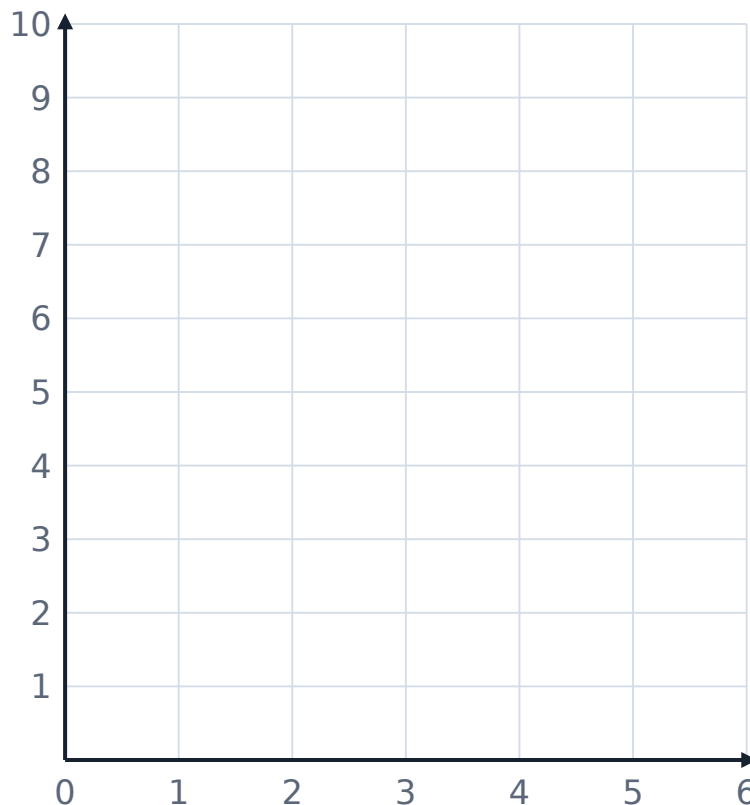
MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A (x): start at 0 and add 1.

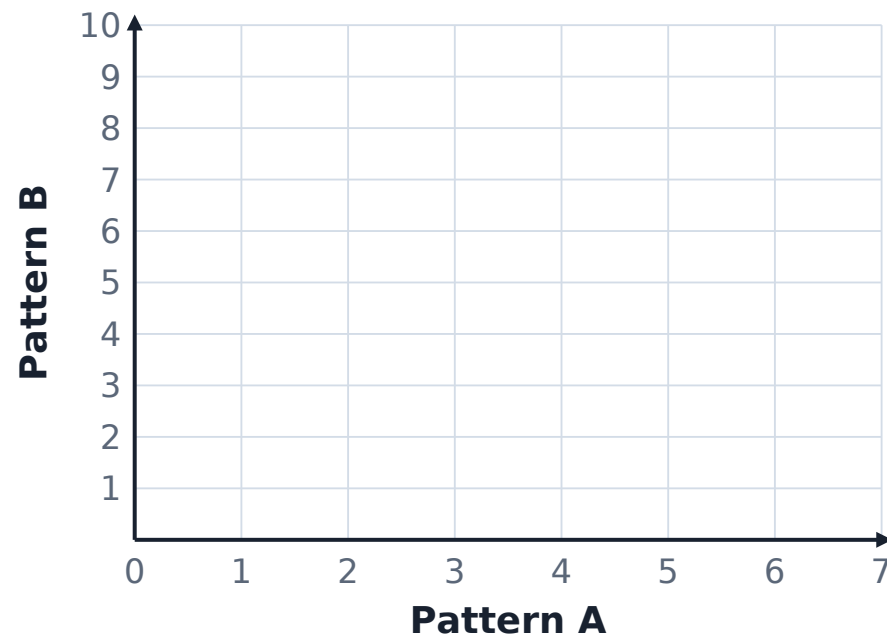
Pattern B (y): start at 0 and add 2.

Graph the ordered pairs for the first 5 terms.



Graph the ordered pairs from the two patterns.

Pattern A	0	2	4	6
Pattern B	0	3	6	9



Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

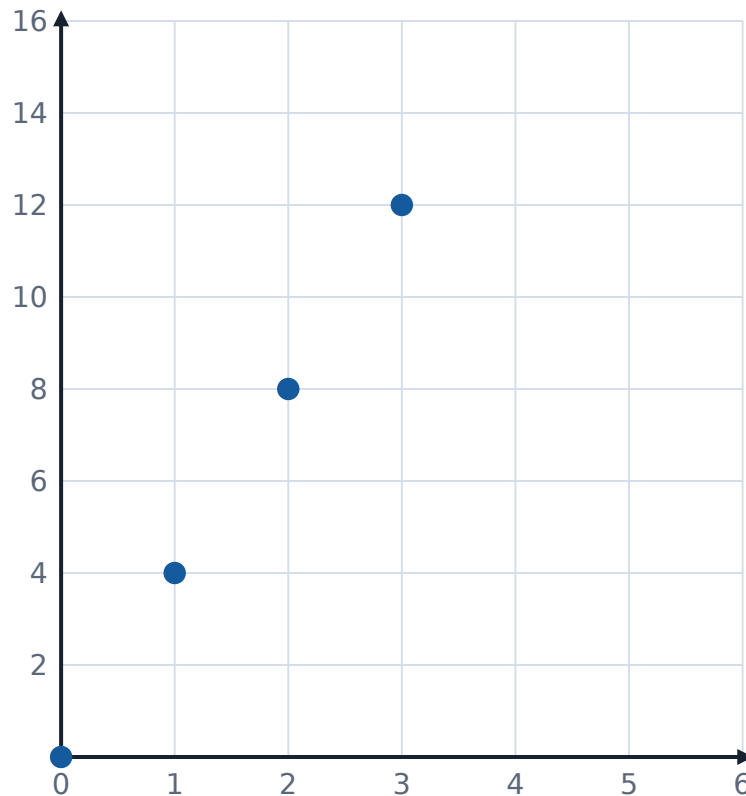
Write the first four ordered pairs (A, B).

Ordered pairs: _____

The points come from two patterns.

Which statement is true?

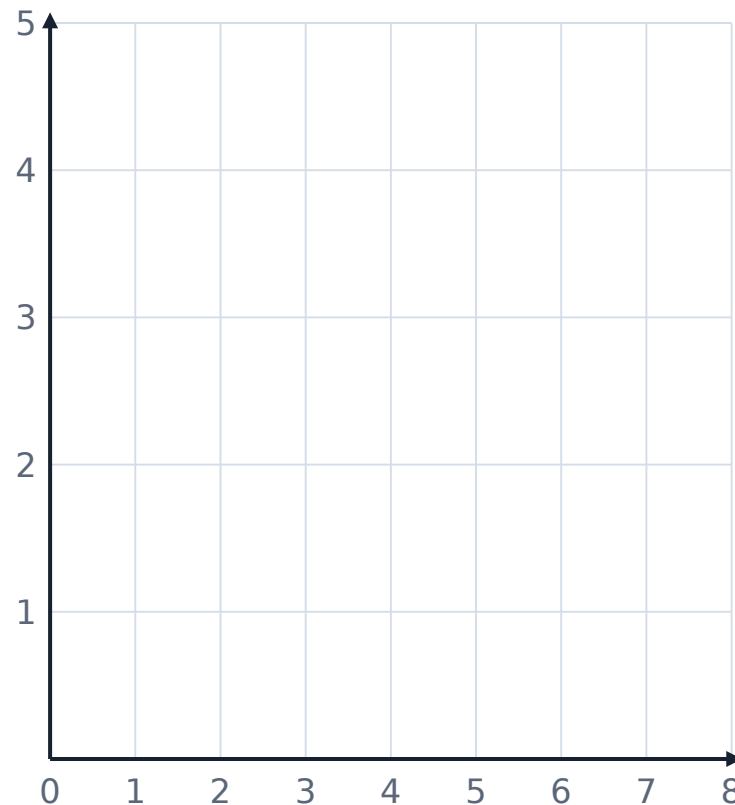
- A.** The y-values increase by 1 each time.
- B.** Each y-value is 4 times its x-value.
- C.** Each x-value is 4 times its y-value.
- D.** Each y-value is 4 more than its x-value.



Pattern A (x): start at 0 and add 2.

Pattern B (y): start at 0 and add 1.

Graph the ordered pairs for the first 4 terms.



SET 7 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 3." The pattern starts at 0.
What is the 5th term?

Term: _____

True or false?

The pattern 0, 5, 10, 15 follows the rule "add 5."

A. True

B. False

Which pattern follows the rule "add 2," starting at 0?

A. 0, 2, 4, 6

B. 2, 4, 6, 8

C. 0, 1, 2, 3

D. 0, 2, 4, 8

What is the rule for the pattern?

0, 8, 16, 24

Rule: _____

The rule is "add 10." The pattern starts at 0.
What is the 6th term?

Term: _____

SET 7 • 5.OA.B.3

BACKWARD 2

Ordered pairs and the coordinate plane • Grade 5 • Geometry • 5.G.A.1

True or false?

In $(3, 6)$, the 3 tells how far to move along the x-axis.

A. True

B. False

In the ordered pair (4, 9), which number is the y-coordinate?

- A.** 9
- B.** 4
- C.** 13
- D.** 5

Which ordered pair names the origin?

Ordered pair: _____

True or false?

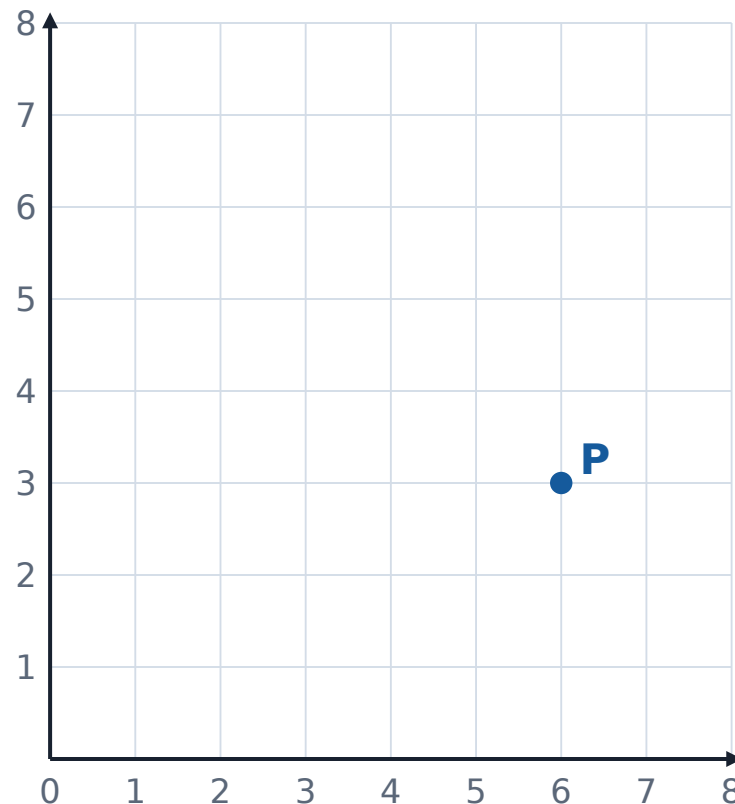
$(2, 5)$ and $(5, 2)$ name the same point.

A. True

B. False

What are the coordinates of point P?

P = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 5.OA.B.3

BACKWARD 3

Reading the scale on an axis • Grade 3 • Measurement & Data • 3.MD.B.3

An axis is numbered 0, 2, 4, 6, 8.

How much does each grid line stand for?

A. 4

B. 1

C. 8

D. 2

True or false?

On an axis numbered 0, 4, 8, 12, the value 10 is halfway between 4 and 8.

- A. True
- B. False

An axis is numbered 0, 5, 10, 15.

Where does 20 go?

Answer: _____

On an axis numbered by 3s, which value is two grid lines above 0?

A. 6

B. 3

C. 9

D. 2

True or false?

On an axis numbered by 10s, the value 25 is on a grid line.

- A.** True
- B.** False

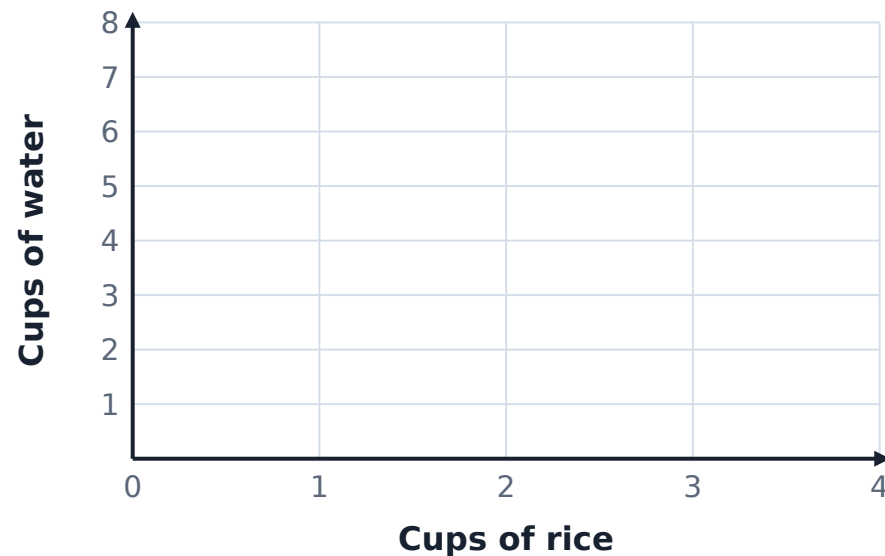
SET 7 • 5.OA.B.3

FORWARD 1

Plot pairs of values from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Plot the pairs of values from the ratio table on the coordinate plane.

Cups of rice	1	2	3
Cups of water	2	4	6



The table shows equivalent ratios.

Find the missing value.

Laps	2	4	6	8
Minutes	5	10	?	20

Missing value: _____

Which point comes from the ratio table?

Boxes	1	2	3
Pens	6	12	18

A. (2, 6)

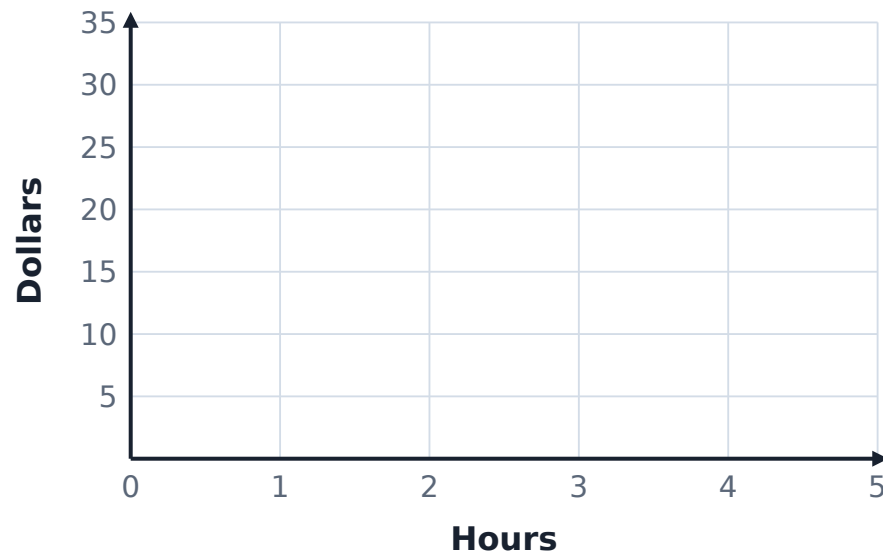
B. (12, 2)

C. (2, 12)

D. (3, 12)

Plot the pairs of values from the ratio table on the coordinate plane.

Hours	1	2	3	4
Dollars	8	16	24	32



True or false?

The points (1, 3), (2, 6), and (3, 9) come from a ratio table with the ratio 1 : 3.

- A. True
- B. False

SET 7 • 5.OA.B.3

FORWARD 2

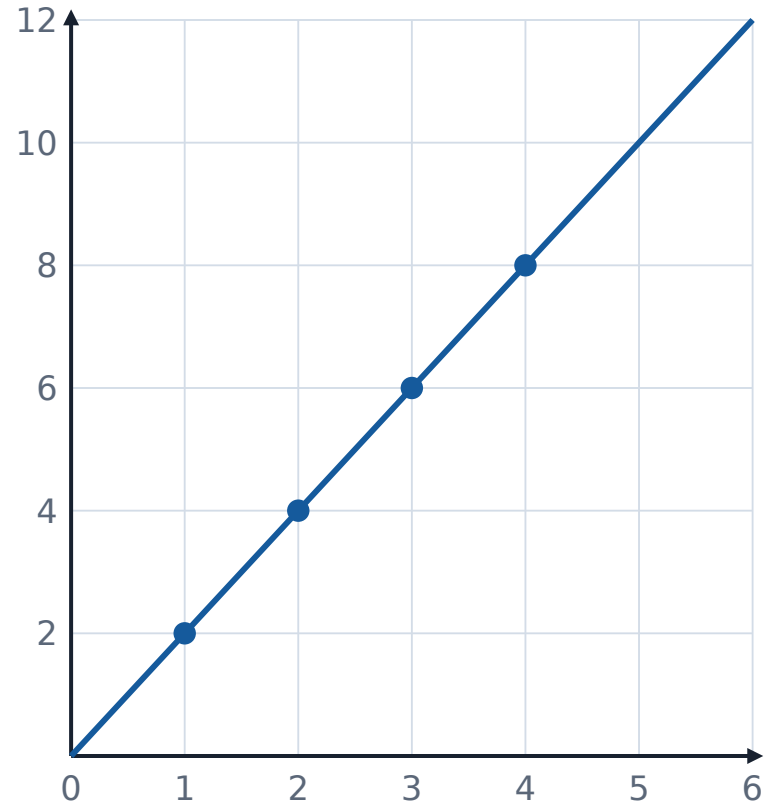
Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The points lie on a straight line through the origin, so the relationship is proportional.

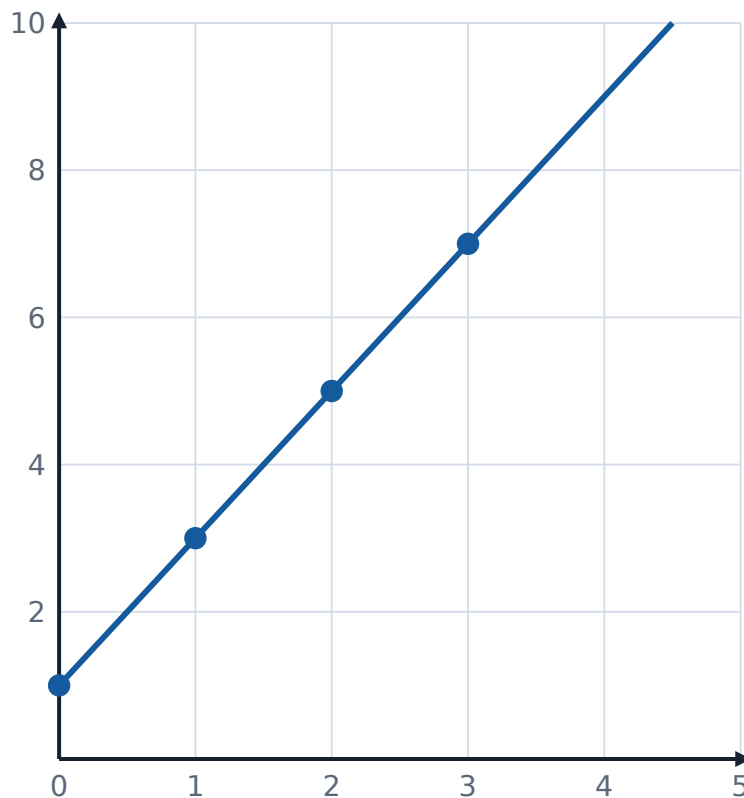
A. True

B. False



Which statement about the graph is true?

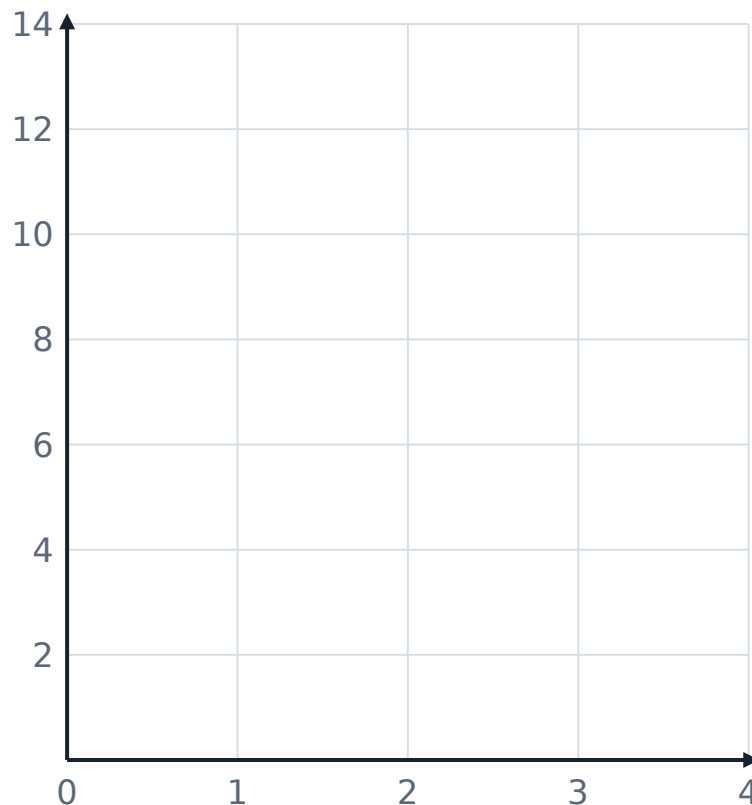
- A.** It is not proportional because the line does not pass through $(0, 0)$.
- B.** It is not proportional because the points are not on a line.
- C.** It is proportional because the points are on a line.
- D.** It is proportional because y increases by 2.



Graph the pairs $(1, 4)$, $(2, 8)$,
and $(3, 12)$.

Is the relationship
proportional?

Answer: _____



True or false?

The points (1, 2), (2, 5), and (3, 8) show a proportional relationship.

- A. True
- B. False

Which set of points shows a proportional relationship?

- A.** (1, 1), (2, 4), (3, 9)
- B.** (1, 2), (2, 3), (3, 4)
- C.** (0, 3), (1, 6), (2, 9)
- D.** (2, 5), (4, 10), (6, 15)

A digit is worth 10 times as much as the same digit one place to its right

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	Each place is worth 10 times the place to its right	4.NBT.A.1
BACKWARD 2	Tenths and hundredths as decimals	4.NF.C.6
BACKWARD 3	Tenths as hundredths	4.NF.C.5
FORWARD 1	Compute with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 8 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

In 5,550, how does the value of the 5 in the hundreds place compare to the value of the 5 in the tens place?

- A.** It is 100 times as much.
- B.** It is 10 times as much.
- C.** It is the same.
- D.** It is $\frac{1}{10}$ as much.

In 3.33, the first 3 is in the ones place and the second 3 is in the tenths place.

How many times as much is the first 3 worth as the second 3?

Times as much: _____

True or false?

In 0.77, the 7 in the tenths place is worth 10 times as much as the 7 in the hundredths place.

- A. True
- B. False

Find the value of the 4 in 642.18 and the value of the 4 in 6,421.8.

How many times as much is the second 4 worth?

4 in 642.18: _____

4 in 6,421.8: _____

Times as much: _____

In which number is the 6 worth 10 times as much as the 6 in 3.6?

A. 0.36

B. 3.06

C. 46.1

D. 60.3

SET 8 • 5.NBT.A.1

BACKWARD 1

Each place is worth 10 times the place to its right • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 880, the 8 in the hundreds place is worth 10 times as much as the 8 in the tens place.

A. True

B. False

What is the value of the 7 in 2,740?

Value: _____

Which number is 10 times as much as 60?

- A.** 600
- B.** 6,000
- C.** 70
- D.** 6

How many tens are in 1 hundred?

Tens: _____

True or false?

70 is 10 times as much as 7.

A. True

B. False

SET 8 • 5.NBT.A.1

BACKWARD 2

Tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{3}{10}$ as a decimal.

Decimal: _____

Write $\frac{47}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.6 = \frac{6}{100}$$

A. True

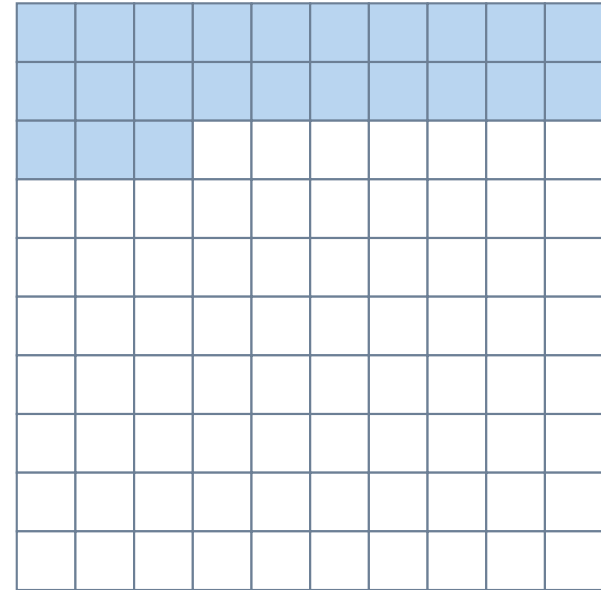
B. False

Which decimal shows 5 hundredths?

- A.** 0.5
- B.** 0.05
- C.** 0.005
- D.** 5.0

What decimal does the shaded part of the grid show?

Decimal: _____



SET 8 • 5.NBT.A.1

BACKWARD 3

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

True or false?

$$\frac{1}{10} = \frac{10}{100}$$

A. True

B. False

How many hundredths are in 1 tenth?

Hundredths: _____

Which fraction is equal to $\frac{4}{10}$?

A. $\frac{4}{100}$

B. $\frac{400}{100}$

C. $\frac{14}{100}$

D. $\frac{40}{100}$

Find the missing number.

$$\frac{7}{10} = \frac{?}{100}$$

Missing number: _____

True or false?

$$\frac{3}{10} + \frac{5}{100} = \frac{35}{100}$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 1

Compute with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$27.6 + 4.38$$

Sum: _____

Subtract.

$$50.2 - 7.65$$

Difference: _____

Multiply.

$$1.2 \times 0.4$$

A. 4.8

B. 0.48

C. 0.048

D. 1.6

Divide.

$$7.5 \div 2.5$$

Quotient: _____

True or false?

$$0.3 \times 0.3 = 0.9$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver at -4.5 m descends 2.25 m.

What is the diver's new position?

Position: _____

The temperature was 3.5°F . It fell 6.2°F .
What is the new temperature?

Temperature: _____

A stock's value changed by $-\$0.75$ each day for 4 days.
What was the total change?

- A.** $-\$0.30$
- B.** $-\$4.75$
- C.** $\$3.00$
- D.** $-\$3.00$

A bank balance of $-\$12.40$ increases by $\$20.00$.

What is the new balance?

Balance: _____

True or false?

$$-1.5 \times 4 = 6$$

A. True

B. False

A digit is worth $\frac{1}{10}$ of the same digit one place to its left

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	One place to the right is worth $\frac{1}{10}$ as much	4.NBT.A.1
BACKWARD 2	The unit fraction $\frac{1}{10}$	3.NF.A.1
BACKWARD 3	Decimals for tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Write fractions as decimals using long division	7.NS.A.2.d

SET 9 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

True or false?

In 44.4, the 4 in the tenths place is worth $\frac{1}{10}$ of the 4 in the ones place.

- A.** True
- B.** False

In 2.22, the 2 in the hundredths place is worth what part of the 2 in the tenths place?

A. 10 times as much

B. $\frac{1}{10}$

C. The same amount

D. $\frac{1}{100}$

What is $\frac{1}{10}$ of 50?

Answer: _____

What is $\frac{1}{10}$ of 0.8?

Answer: _____

In which number is the 9 worth $\frac{1}{10}$ of the 9 in 19.4?

- A.** 5.92
- B.** 3.009
- C.** 9.01
- D.** 90.1

SET 9 • 5.NBT.A.1

BACKWARD 1

One place to the right is worth $\frac{1}{10}$ as much • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

True or false?

In 330, the 3 in the tens place is worth $\frac{1}{10}$ of the 3 in the hundreds place.

- A.** True
- B.** False

Divide.

$$400 \div 10$$

Quotient: _____

Which number is $\frac{1}{10}$ of 3,000?

- A.** 3
- B.** 300
- C.** 30,000
- D.** 30

How many hundreds are in 1 thousand?

Hundreds: _____

True or false?

$$60 \div 10 = 600$$

A. True

B. False

SET 9 • 5.NBT.A.1

BACKWARD 2

The unit fraction $\frac{1}{10}$ • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{10}$ means 1 of 10 equal parts.

A. True

B. False

A bar is cut into 10 equal parts.

What fraction of the bar is one part?

Fraction: _____

Which fraction is greater?

A. $\frac{1}{10}$

B. $\frac{1}{2}$

True or false?

A whole cut into 10 equal parts has parts that are each $\frac{1}{10}$ of the whole.

- A.** True
- B.** False

What fraction of the bar is shaded?



Fraction: _____

SET 9 • 5.NBT.A.1

BACKWARD 3

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{9}{100}$ as a decimal.

Decimal: _____

Write 0.8 as a fraction.

Fraction: _____

True or false?

0.4 and 0.04 have the same value.

A. True

B. False

Which decimal is equal to $\frac{6}{10}$?

A. 6.0

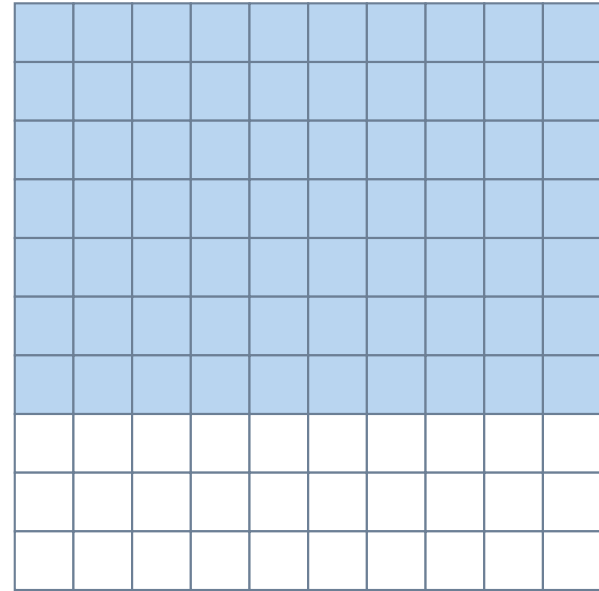
B. 0.06

C. 6.10

D. 0.6

What decimal does the shaded part of the grid show?

Decimal: _____



SET 9 • 5.NBT.A.1

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$4.8 \div 0.4$$

Quotient: _____

Divide.

$$9.36 \div 3$$

Quotient: _____

Divide.

$$0.56 \div 0.7$$

A. 80

B. 0.08

C. 8

D. 0.8

Divide.

$$15 \div 0.25$$

Quotient: _____

True or false?

$$6.3 \div 0.9 = 0.7$$

A. True

B. False

SET 9 • 5.NBT.A.1

FORWARD 2

Write fractions as decimals using long division • Grade 7 • The Number System
• 7.NS.A.2.d

Use long division to write $\frac{7}{8}$ as a decimal.

Decimal: _____

Use long division to write $\frac{2}{9}$ as a decimal.

Decimal: _____

Which fraction has a terminating decimal?

A. $\frac{5}{6}$

B. $\frac{4}{7}$

C. $\frac{1}{3}$

D. $\frac{3}{20}$

Use long division to write $\frac{9}{20}$ as a decimal.

Decimal: _____

True or false?

The decimal form of $\frac{1}{11}$ repeats.

A. True

B. False

Multiply and divide whole numbers by powers of 10; write powers of 10 with exponents

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Multiplying and dividing by 10	4.NBT.A.1
BACKWARD 2	Multiplying by multiples of 10	3.NBT.A.3
FORWARD 1	Write and evaluate whole-number exponents	6.EE.A.1
FORWARD 2	Multiply and divide negative numbers by powers of 10 — nearest...	7.NS.A.2.c

SET 10 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$37 \times 1,000$$

Product: _____

Divide.

$$45,000 \div 100$$

Quotient: _____

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

$$6 \times 10^3 = ?$$

A. 60,000

B. 6,000

C. 18

D. 600

True or false?

$$820 \div 10^2 = 82$$

A. True

B. False

SET 10 • 5.NBT.A.2

BACKWARD 1

Multiplying and dividing by 10 • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Multiply.

$$25 \times 10$$

Product: _____

True or false?

300 is 10 times as much as 30.

A. True

B. False

Which number is 10 times as much as 450?

- A.** 4,500
- B.** 45,000
- C.** 460
- D.** 45

How many tens are in 800?

Tens: _____

Divide.

$$7,000 \div 10$$

Quotient: _____

SET 10 • 5.NBT.A.2

BACKWARD 2

Multiplying by multiples of 10 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

Multiply.

$$4 \times 60$$

Product: _____

Multiply.

$$9 \times 80$$

Product: _____

True or false?

$$7 \times 30 = 2,100$$

A. True

B. False

$$5 \times 90 = ?$$

A. 4,500

B. 450

C. 45

D. 95

Multiply.

$$8 \times 70$$

Product: _____

SET 10 • 5.NBT.A.2

FORWARD 1

Write and evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$3^4$$

Value: _____

Write $5 \times 5 \times 5$ using an exponent.

Exponent form: _____

$$2^5 = ?$$

A. 25

B. 10

C. 64

D. 32

Evaluate.

$$10^3 + 2^2$$

Value: _____

True or false?

$$4^3 = 12$$

A. True

B. False

FORWARD 2

Multiply and divide negative numbers by powers of 10 • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Multiply.

$$-3.5 \times 100$$

Product: _____

Divide.

$$-420 \div 10$$

Quotient: _____

$$-0.06 \times 1,000 = ?$$

- A.** -60
- B.** -0.006
- C.** -6
- D.** 60

Divide.

$$7.2 \div (-10)$$

Quotient: _____

True or false?

$$(-10) \times (-10) \times (-10) = -1,000$$

A. True

B. False

Place the decimal point when multiplying or dividing a decimal by a power of 10

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Decimal notation for fractions	4.NF.C.6
BACKWARD 2	Place value when multiplying by 10 or 100	4.NBT.A.1
FORWARD 1	Compute fluently with multi-digit decimals	6.NS.B.3
FORWARD 2	Powers of 10 with negative rational numbers — nearest related (n...	7.NS.A.2.c

SET 11 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$3.82 \times 100$$

Product: _____

Divide.

$$56.4 \div 10$$

Quotient: _____

Multiply.

$$0.075 \times 10^2$$

Product: _____

Divide.

$$2.9 \div 10^3$$

A. 0.029

B. 0.29

C. 0.0029

D. 2,900

True or false?

When you multiply a number by 1,000, each digit moves 3 places to the left.

- A. True
- B. False

SET 11 • 5.NBT.A.2

BACKWARD 1

Decimal notation for fractions • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

Write $\frac{35}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.7 = \frac{7}{100}$$

A. True

B. False

Which decimal is $2\frac{4}{10}$?

A. 24

B. 2.4

C. 0.24

D. 2.04

Write 0.09 as a fraction.

Fraction: _____

True or false?

$$1.5 = 1\frac{5}{100}$$

A. True

B. False

SET 11 • 5.NBT.A.2

BACKWARD 2

Place value when multiplying by 10 or 100 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

When a whole number is multiplied by 10, each digit moves one place to the left.

A. True

B. False

Multiply.

$$64 \times 100$$

Product: _____

$$3,500 \div 10 = ?$$

- A.** 350
- B.** 3,510
- C.** 35
- D.** 35,000

Divide.

$$9,000 \div 100$$

Quotient: _____

True or false?

$$500 \div 10 = 5$$

A. True

B. False

SET 11 • 5.NBT.A.2

FORWARD 1

Compute fluently with multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

Multiply.

$$2.4 \times 0.35$$

Product: _____

Divide.

$$1.44 \div 0.12$$

Quotient: _____

Multiply.

$$12.5 \times 0.8$$

A. 100

B. 10

C. 0.1

D. 1

Add.

$$0.875 + 12.4$$

Sum: _____

True or false?

$$4.5 \div 0.05 = 9$$

A. True

B. False

FORWARD 2

Powers of 10 with negative rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Find the value.

$$-0.45 \times 100$$

Value: _____

Find the value.

$$-63.5 \div 1,000$$

Value: _____

Find the value.

$$(-2.7) \times (-10)$$

A. -0.27

B. -27

C. 0.27

D. 27

Find the value.

$$0.8 \times (-100)$$

Value: _____

True or false?

$$-5.2 \div 100 = -0.052$$

A. True

B. False

Read and write decimals to thousandths using numerals and words

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Decimals for tenths and hundredths	4.NF.C.6
BACKWARD 2	Reading and writing whole numbers	4.NBT.A.2
BACKWARD 3	Naming tenths and hundredths	4.NF.C.5
FORWARD 1	Find and position rational numbers on a number line	6.NS.C.6.c
FORWARD 2	Write fractions as decimals	7.NS.A.2.d

SET 12 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write the number in standard form.

four and twenty-six hundredths

Number: _____

Write 3.508 in words.

Words: _____

Which number is nine and forty-one thousandths?

- A.** 9.410
- B.** 9.41
- C.** 9.0041
- D.** 9.041

Write the number in standard form.

seventeen and three tenths

Number: _____

True or false?

0.072 is read "seventy-two hundredths."

A. True

B. False

SET 12 • 5.NBT.A.3.a

BACKWARD 1

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.5 = \frac{5}{10}$$

A. True

B. False

Which decimal is equal to $\frac{12}{100}$?

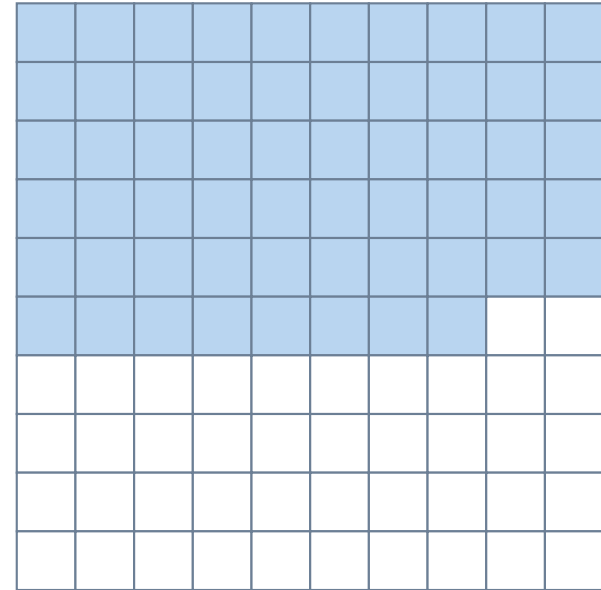
- A.** 1.2
- B.** 12.0
- C.** 0.12
- D.** 0.012

Write $3\frac{7}{10}$ as a decimal.

Decimal: _____

What decimal does the shaded part of the grid show?

Decimal: _____



SET 12 • 5.NBT.A.3.a

BACKWARD 2

Reading and writing whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write the number in standard form.

two thousand, four hundred six

Number: _____

True or false?

5,030 is read "five thousand three hundred."

A. True

B. False

Which number is eighty thousand, nine hundred?

A. 80,900

B. 8,900

C. 89,000

D. 80,090

Write 607 in words.

Words: _____

Write the number in standard form.

twelve thousand, fifty

Number: _____

SET 12 • 5.NBT.A.3.a

BACKWARD 3

Naming tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

True or false?

Three tenths is equal to thirty hundredths.

A. True

B. False

How many hundredths are in 4 tenths?

Hundredths: _____

Which fraction is "five hundredths"?

A. $\frac{100}{5}$

B. $\frac{5}{10}$

C. $\frac{50}{100}$

D. $\frac{5}{100}$

Write "nine tenths" as a fraction.

Fraction: _____

True or false?

$$\frac{60}{100} = \frac{6}{10}$$

A. True

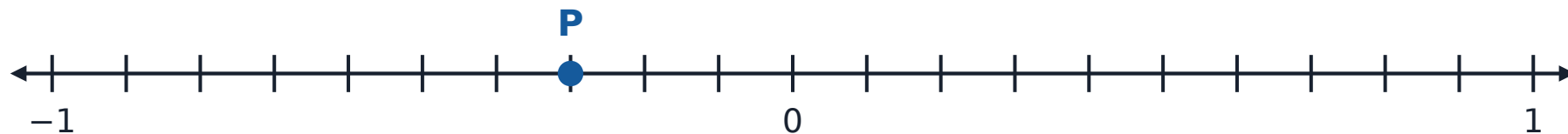
B. False

SET 12 • 5.NBT.A.3.a

FORWARD 1

Find and position rational numbers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?

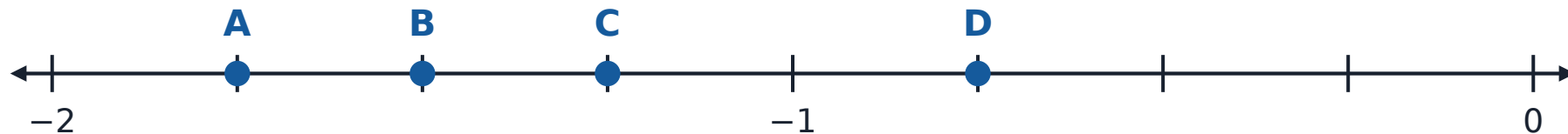


P = _____

Plot and label point Q at 0.45 on the number line.



Which point is at -1.25 ?



A. Point A

B. Point B

C. Point C

D. Point D

What number does point R represent?



R = _____

True or false?

On a number line, -0.6 is to the left of -0.5 .

A. True

B. False

SET 12 • 5.NBT.A.3.a

FORWARD 2

Write fractions as decimals • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

Use long division to write $\frac{5}{12}$ as a decimal.

Decimal: _____

$$\frac{11}{20} = ?$$

A. 0.55

B. 5.5

C. 0.2

D. 0.11

Use long division to write $\frac{4}{9}$ as a decimal.

Decimal: _____

True or false?

$$\frac{7}{25} = 0.32$$

A. True

B. False

Write decimals to thousandths in expanded form

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Expanded form of whole numbers	4.NBT.A.2
BACKWARD 2	Decimals and fractions with denominators 10 and 100	4.NF.C.6
FORWARD 1	Write expanded form with exponents	6.EE.A.1
FORWARD 2	Decimal expansions of fractions — nearest related (no direct Gra...	7.NS.A.2.d

SET 13 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write 352.68 in expanded form.

Expanded form: _____

Write the number in standard form.

$$4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 9 \times \frac{1}{1000}$$

Number: _____

Which is the expanded form of 6.045?

A. $6 + 45$

B. $6 \times 10 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

C. $6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

D. $6 \times 1 + 4 \times \frac{1}{10} + 5 \times \frac{1}{100}$

Write the number in standard form.

$$8 \times 100 + 2 \times \frac{1}{10} + 6 \times \frac{1}{100}$$

Number: _____

True or false?

$$0.908 = 9 \times \frac{1}{10} + 8 \times \frac{1}{1000}$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 1

Expanded form of whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write 4,371 in expanded form.

Expanded form: _____

True or false?

$$6,052 = 6,000 + 500 + 2$$

A. True

B. False

Which number is $7,000 + 400 + 9$?

- A.** 7,409
- B.** 7,490
- C.** 74,009
- D.** 7,049

Write $3 \times 1,000 + 8 \times 100 + 5 \times 1$ in standard form.

Number: _____

True or false?

$$900 + 20 + 4 = 9,024$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 2

Decimals and fractions with denominators 10 and 100 • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 0.6 as a fraction.

Fraction: _____

True or false?

$$\frac{4}{100} = 0.4$$

A. True

B. False

Which fraction is equal to 0.25?

A. $\frac{1}{25}$

B. $\frac{25}{100}$

C. $\frac{25}{10}$

D. $\frac{2}{5}$

Write $2 + \frac{3}{10} + \frac{7}{100}$ as a decimal.

Decimal: _____

True or false?

$$\frac{9}{10} = 0.09$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

FORWARD 1

Write expanded form with exponents • Grade 6 • Expressions & Equations •
6.EE.A.1

Write $5,000 + 300 + 20 + 7$ using powers of 10.

Expression: _____

Evaluate.

$$4 \times 10^3 + 6 \times 10^1$$

Value: _____

Which number is equal to 10^4 ?

- A.** 40
- B.** 100,000
- C.** 1,000
- D.** 10,000

Write $600 + 90 + 1$ using powers of 10.

Expression: _____

True or false?

$$2 \times 10^2 + 5 = 250$$

A. True

B. False

FORWARD 2

Decimal expansions of fractions • Grade 7 • The Number System • 7.NS.A.2.d

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Write $\frac{3}{4}$ as a decimal.

Decimal: _____

Use long division to write $\frac{1}{6}$ as a decimal.

Decimal: _____

Which fraction is equal to 0.666... (the 6 repeats)?

A. $\frac{3}{5}$

B. $\frac{6}{10}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

Use long division to write $\frac{5}{8}$ as a decimal.

Decimal: _____

True or false?

$$\frac{2}{5} = 0.4$$

A. True

B. False

Compare decimals to thousandths

MAIN	Grade 5 main question	5.NBT.A.3.b
BACKWARD 1	Comparing decimals to hundredths	4.NF.C.7
BACKWARD 2	The symbols $>$, $<$, and $=$	1.NBT.B.3
FORWARD 1	Order rational numbers	6.NS.C.7.b
FORWARD 2	Compare fractions and decimals using decimal expansions	7.NS.A.2.d

SET 14 • 5.NBT.A.3.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Write $>$, $<$, or $=$ to compare.

0.47 0.405

Symbol: _____

Write $>$, $<$, or $=$ to compare.

3.06 ____ 3.060

Symbol: _____

Which number is greatest?

- A.** 5.2
- B.** 5.19
- C.** 5.093
- D.** 5.199

Order from least to greatest.

2.31, 2.301, 2.13, 2.3

Order: _____

True or false?

$$0.8 < 0.79$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 1

Comparing decimals to hundredths • Grade 4 • Number & Operations–Fractions
• 4.NF.C.7

Write $>$, $<$, or $=$ to compare.

0.6 0.58

Symbol: _____

True or false?

$$0.3 = 0.03$$

A. True

B. False

Which number is less?

A. 0.09

B. 0.1

Write $>$, $<$, or $=$ to compare.

0.75 0.8

Symbol: _____

True or false?

$$0.42 > 0.5$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 2

The symbols $>$, $<$, and $=$ • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

True or false?

$$9 > 6$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

52 25

Symbol: _____

True or false?

The symbol $>$ means "is less than."

- A. True
- B. False

Which statement is true?

A. $14 < 41$

B. $14 = 41$

C. $14 > 41$

SET 14 • 5.NBT.A.3.b

FORWARD 1

Order rational numbers • Grade 6 • The Number System • 6.NS.C.7.b

Write $>$, $<$, or $=$ to compare.

-0.4 ____ -0.45

Symbol: _____

Which number is least?

A. -2

B. -2.5

C. 0.5

D. -2.05

Order from least to greatest.

-1.2 , 0.3 , -1.25 , 0

Order: _____

True or false?

$$-3.7 > -3.6$$

A. True

B. False

A temperature of -2.5°C is colder than a temperature of -1.8°C .

Write an inequality to show this.

Inequality: _____

SET 14 • 5.NBT.A.3.b

FORWARD 2

Compare fractions and decimals using decimal expansions • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Is $\frac{5}{8}$ greater than or less than 0.6?

Decimal: _____

Answer: _____

Which number is greatest?

A. $\frac{4}{5}$

B. 0.7

C. $\frac{3}{4}$

D. 0.78

Order from least to greatest.

$$\frac{2}{3}, 0.6, \frac{5}{8}$$

Order: _____

True or false?

$$\frac{1}{3} > 0.33$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

$$\frac{7}{20} \text{ ____ } 0.4$$

Symbol: _____

Round decimals to any place

MAIN

Grade 5 main question

5.NBT.A.4

BACKWARD 1

Rounding whole numbers

4.NBT.A.3

BACKWARD 2

Decimals on a number line

4.NF.C.6

FORWARD 1

Estimate to check computations with decimals

6.NS.B.3

FORWARD 2

Use estimation to judge whether an answer is reasonable

7.EE.B.3

SET 15 • 5.NBT.A.4

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 6.483 to the nearest tenth.

Rounded: _____

Round 12.749 to the nearest hundredth.

Rounded: _____

Round 0.865 to the nearest whole number.

A. 0.9

B. 1

C. 0

D. 0.87

Round 39.96 to the nearest tenth.

Rounded: _____

True or false?

4.351 rounded to the nearest hundredth is 4.36.

A. True

B. False

SET 15 • 5.NBT.A.4

BACKWARD 1

Rounding whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 4,682 to the nearest hundred.

Rounded: _____

True or false?

350 rounded to the nearest hundred is 300.

A. True

B. False

Round 7,249 to the nearest thousand.

- A.** 7,200
- B.** 7,000
- C.** 7,250
- D.** 8,000

Round 85 to the nearest ten.

Rounded: _____

True or false?

1,349 rounded to the nearest hundred is 1,400.

A. True

B. False

SET 15 • 5.NBT.A.4

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal is at point A?



A = _____

True or false?

0.7 is closer to 1 than to 0.

A. True

B. False

Is 0.43 closer to 0.4 or to 0.5?

A. 0.4

B. 0.5

What decimal is at point B?



B = _____

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

SET 15 • 5.NBT.A.4

FORWARD 1

Estimate to check computations with decimals • Grade 6 • The Number System
• 6.NS.B.3

Round each number to the nearest whole number. Then add to estimate.

$$18.7 + 6.25$$

Estimate: _____

Which is the best estimate of 4.9×6.1 ?

A. 36

B. 30

C. 24

D. 11

Estimate $31.8 \div 7.9$ by rounding each number to the nearest whole number.

Estimate: _____

True or false?

29.6 – 10.2 is about 40.

A. True

B. False

Kim says $3.9 \times 2.1 = 81.9$.

Use an estimate to decide whether her answer is reasonable. Write yes or no.

Answer: _____

SET 15 • 5.NBT.A.4

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A \$24.75 shirt is 20% off.

Estimate the sale price.

Estimate: _____

A worker earns \$19.85 per hour for 9.75 hours.
Which is the best estimate of her pay?

- A.** \$20
- B.** \$2,000
- C.** \$200
- D.** \$100

True or false?

A 6.2% tax on \$49.50 is about \$3.

A. True

B. False

Nina says $\frac{3}{4} \times 39.8$ is about 40.

Is her estimate reasonable? Give a better estimate.

Reasonable? _____

Better estimate: _____

A board is $8\frac{1}{4}$ feet long. It is cut into pieces that are $1\frac{7}{8}$ feet long.

Estimate how many whole pieces can be cut.

Estimate: _____

Multiply multi-digit whole numbers using the standard algorithm

MAIN	Grade 5 main question	5.NBT.B.5
BACKWARD 1	Multiplying by a one-digit or two-digit number	4.NBT.B.5
BACKWARD 2	Multiplication facts	3.OA.C.7
BACKWARD 3	Adding partial products	4.NBT.B.4
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply integers	7.NS.A.2.c

SET 16 • 5.NBT.B.5

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Multiply.

$$246 \times 37$$

Product: _____

Multiply.

$$1,385 \times 24$$

Product: _____

Multiply.

$$508 \times 63$$

- A.** 3,556
- B.** 32,104
- C.** 32,004
- D.** 31,904

Multiply.

$$79 \times 86$$

Product: _____

A theater has 48 rows with 125 seats in each row.
How many seats are in the theater?

Seats: _____

SET 16 • 5.NBT.B.5

BACKWARD 1

Multiplying by a one-digit or two-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$346 \times 7$$

Product: _____

Multiply.

$$23 \times 45$$

Product: _____

True or false?

$$1,203 \times 4 = 4,082$$

A. True

B. False

$$62 \times 18 = ?$$

A. 1,126

B. 496

C. 1,116

D. 1,006

Multiply.

$$58 \times 30$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

Multiply.

$$8 \times 9$$

Product: _____

True or false?

$$6 \times 9 = 56$$

A. True

B. False

$$4 \times 8 = ?$$

A. 24

B. 36

C. 32

D. 28

Multiply.

$$9 \times 9$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 3

Adding partial products • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$7,380 + 1,722$$

Sum: _____

Add.

$$27,700 + 5,540$$

Sum: _____

True or false?

$$30,480 + 1,524 = 31,904$$

A. True

B. False

$$6,400 + 394 = ?$$

A. 6,784

B. 6,794

C. 7,794

D. 6,694

Add.

$$4,800 + 1,200$$

Sum: _____

SET 16 • 5.NBT.B.5

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$2.46 \times 3.7$$

Product: _____

Multiply.

$$13.85 \times 2.4$$

Product: _____

Multiply.

$$5.08 \times 6.3$$

A. 31.904

B. 320.04

C. 3.2004

D. 32.004

Multiply.

$$0.79 \times 8.6$$

Product: _____

True or false?

$$1.25 \times 4.8 = 60$$

A. True

B. False

SET 16 • 5.NBT.B.5

FORWARD 2

Multiply integers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-24) \times 15$$

Product: _____

Find the product.

$$(-12) \times (-35)$$

Product: _____

Find the product.

$$18 \times (-21)$$

- A.** 378
- B.** -378
- C.** -39
- D.** -388

Find the product.

$$(-3) \times (-4) \times (-25)$$

Product: _____

True or false?

$$(-6)(-7) = -42$$

A. True

B. False

Divide with up to four-digit dividends and two-digit divisors

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Multiplying to check a quotient	4.NBT.B.5
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Divide multi-digit numbers using the standard algorithm	6.NS.B.2
FORWARD 2	Divide integers	7.NS.A.2.b

SET 17 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$2,016 \div 24$$

Quotient: _____

Divide.

$$7,830 \div 45$$

Quotient: _____

Divide.

$$952 \div 17$$

A. 54

B. 46

C. 56

D. 57

Divide.

$$3,724 \div 38$$

Quotient: _____

Divide. Write the quotient and the remainder.

$$1,000 \div 32$$

Quotient and remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 6$$

Quotient: _____

Divide.

$$1,935 \div 5$$

Quotient: _____

True or false?

$$728 \div 8 = 901$$

A. True

B. False

$$504 \div 7 = ?$$

A. 82

B. 71

C. 62

D. 72

Divide. Write the quotient and the remainder.

$$95 \div 4$$

Quotient and remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 2

Multiplying to check a quotient • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

True or false?

To check $2,016 \div 24 = 84$, you can multiply 24×84 .

- A. True
- B. False

Multiply.

$$45 \times 174$$

Product: _____

Which product checks $952 \div 17 = 56$?

- A.** $17 + 56$
- B.** 17×56
- C.** $56 - 17$
- D.** 952×17

Multiply.

$$38 \times 98$$

Product: _____

True or false?

$$32 \times 31 + 8 = 1,000$$

A. True

B. False

SET 17 • 5.NBT.B.6

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$72 \div 8$$

Quotient: _____

Divide.

$$54 \div 9$$

Quotient: _____

True or false?

$$63 \div 7 = 8$$

A. True

B. False

$$40 \div 5 = ?$$

A. 6

B. 7

C. 8

D. 9

Divide.

$$35 \div 7$$

Quotient: _____

SET 17 • 5.NBT.B.6

FORWARD 1

Divide multi-digit numbers using the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$6,432 \div 48$$

Quotient: _____

Use the standard algorithm to divide.

$$25,875 \div 69$$

Quotient: _____

Use the standard algorithm to divide.

$$14,040 \div 52$$

- A.** 207
- B.** 2,700
- C.** 270
- D.** 27

Use the standard algorithm to divide.

$$9,594 \div 41$$

Quotient: _____

True or false?

$$11,700 \div 45 = 2,600$$

A. True

B. False

SET 17 • 5.NBT.B.6

FORWARD 2

Divide integers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-2,016 \div 24$$

Quotient: _____

Find the quotient.

$$(-952) \div (-17)$$

Quotient: _____

Find the quotient.

$$780 \div (-15)$$

- A.** -52
- B.** -53
- C.** -765
- D.** 52

Find the quotient.

$$-(144 \div 12)$$

Quotient: _____

True or false?

$$(-90) \div 18 = 90 \div (-18)$$

A. True

B. False

Explain division using area models and equations

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Area as length times width	3.MD.C.7.b
BACKWARD 3	Splitting a rectangle to find its area	3.MD.C.7.c
FORWARD 1	Use the distributive property with the greatest common factor	6.NS.B.4
FORWARD 2	Factor and expand linear expressions	7.EE.A.1

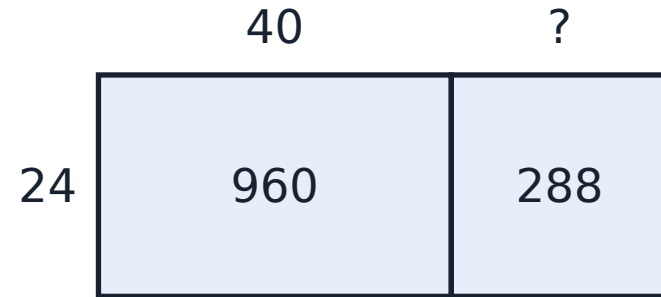
SET 18 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

The area model shows
 $1,248 \div 24$.

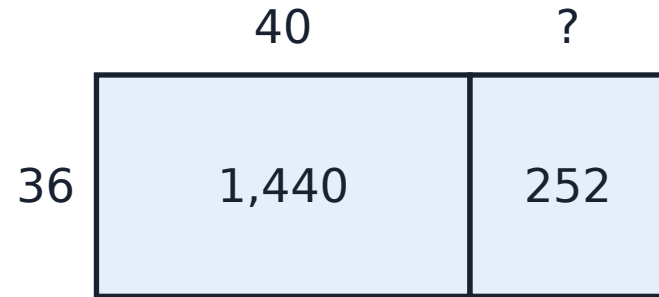
What is the quotient?



Quotient: _____

The area model shows
 $1,692 \div 36$.

What is the quotient?



Quotient: _____

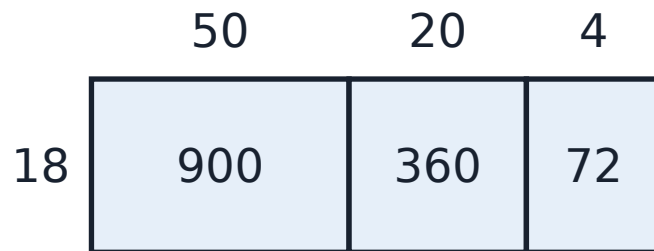
Which equation matches the area model?

A. $1,332 \div 74 = 20$

B. $1,332 \div 18 = 74$

C. $18 \times 50 = 1,332$

D. $900 \div 18 = 74$



Use an area model or equations to divide.

$$2,184 \div 28$$

Quotient: _____

True or false?

$$1,248 \div 24 = (960 \div 24) + (288 \div 24)$$

A. True

B. False

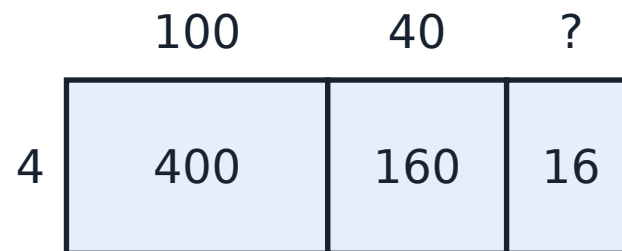
SET 18 • 5.NBT.B.6

BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows
 $576 \div 4$.

What is the quotient?



Quotient: _____

True or false?

$$468 \div 3 = (300 \div 3) + (150 \div 3) + (18 \div 3)$$

A. True

B. False

Which shows $245 \div 5$ split into easier parts?

- A.** $(200 \div 5) + (45 \div 5)$
- B.** $(245 \div 2) + (245 \div 3)$
- C.** $200 \div (5 + 45)$
- D.** $(200 \div 5) \times (45 \div 5)$

Divide.

$$735 \div 7$$

Quotient: _____

A rectangle has an area of 312 square units and a width of 6 units.

What is its length?

Length: _____

SET 18 • 5.NBT.B.6

BACKWARD 2

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 9 units long and 7 units wide.
What is its area?

Area: _____

True or false?

A rectangle with an area of 40 square units and a side of 5 units has another side of 8 units.

A. True

B. False

Which equation gives the area of a 6-by-8 rectangle?

A. $8 - 6 = 2$

B. $6 + 8 = 14$

C. $6 \times 8 = 48$

D. $2 \times (6 + 8) = 28$

A rectangle has an area of 36 square units. One side is 4 units long.

How long is the other side?

Length: _____

Find the area of the rectangle.

Area: _____



SET 18 • 5.NBT.B.6

BACKWARD 3

Splitting a rectangle to find its area • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

The area of a rectangle that is 4 by $(10 + 3)$ equals $4 \times 10 + 4 \times 3$.

A. True

B. False

A 5-by-12 rectangle is split into a 5-by-10 rectangle and a 5-by-2 rectangle.

What is the total area?

Area: _____

A 7-by-13 rectangle is split into a 7-by-10 part and a 7-by-3 part.

Which expression gives its area?

A. $7 \times 10 + 7 \times 3$

B. $7 \times 10 + 3$

C. $7 + 10 + 3$

D. $7 \times 10 \times 3$

Fill in the blank.

$$6 \times 14 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

True or false?

$$8 \times 12 = 8 \times 10 + 2$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 1

Use the distributive property with the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $36 + 48$ as a product.

Expression: _____

Find the greatest common factor of 32 and 56.

GCF: _____

Which expression is equal to $45 + 60$ and uses the greatest common factor?

- A.** $15(3 + 6)$
- B.** $15(3 + 4)$
- C.** $3(15 + 20)$
- D.** $5(9 + 12)$

Complete the equation.

$$24 + 40 = 8(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

True or false?

$$18 + 27 = 9(2 + 3)$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 2

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor using the greatest common factor.

$$12x + 30$$

Expression: _____

Expand.

$$7(4y - 3)$$

Expression: _____

Which expression is equivalent to $18a - 24$?

- A.** $3(6a - 4)$
- B.** $6(3a - 24)$
- C.** $18(a - 6)$
- D.** $6(3a - 4)$

Factor out -4 .

$$-8n - 20$$

Expression: _____

True or false?

$$4(2.5x + 3) = 10x + 3$$

A. True

B. False

Add and subtract decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1

Adding and subtracting multi-digit whole numbers

4.NBT.B.4

BACKWARD 2

Adding tenths and hundredths

4.NF.C.5

BACKWARD 3

Writing decimals with the same number of places

4.NF.C.6

FORWARD 1

Add and subtract multi-digit decimals

6.NS.B.3

FORWARD 2

Add and subtract rational numbers

7.NS.A.1.d

SET 19 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$23.45 + 8.7$$

Sum: _____

Subtract.

$$15.3 - 6.48$$

Difference: _____

Add.

$$4.06 + 12.9$$

- A.** 16.15
- B.** 16.069
- C.** 5.35
- D.** 16.96

Subtract.

$$20 - 7.35$$

Difference: _____

Maya had \$18.50. She spent \$6.75.

How much money does she have left?

Money left: _____

SET 19 • 5.NBT.B.7

BACKWARD 1

Adding and subtracting multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$2,345 + 870$$

Sum: _____

Subtract.

$$1,530 - 648$$

Difference: _____

True or false?

$$406 + 1,290 = 1,596$$

A. True

B. False

$$2,000 - 735 = ?$$

A. 2,735

B. 1,265

C. 1,275

D. 1,365

Subtract.

$$1,850 - 675$$

Difference: _____

SET 19 • 5.NBT.B.7

BACKWARD 2

Adding tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

Add.

$$\frac{3}{10} + \frac{45}{100}$$

Sum: _____

True or false?

$$\frac{6}{10} + \frac{2}{100} = \frac{8}{100}$$

A. True

B. False

$$\frac{7}{10} + \frac{15}{100} = ?$$

A. $\frac{85}{100}$

B. $\frac{22}{100}$

C. $\frac{22}{110}$

D. $\frac{85}{10}$

Write $\frac{4}{10}$ as hundredths.

Fraction: _____

True or false?

$$\frac{5}{10} + \frac{5}{100} = \frac{10}{100}$$

A. True

B. False

SET 19 • 5.NBT.B.7

BACKWARD 3

Writing decimals with the same number of places • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

True or false?

$$8.7 = 8.70$$

A. True

B. False

Write 15.3 with two decimal places.

Decimal: _____

Which decimal is equal to 20?

- A.** 20.00
- B.** 0.20
- C.** 2.00
- D.** 200.0

True or false?

$$6.4 = 6.04$$

A. True

B. False

Write 12.9 with a digit in the hundredths place, without changing its value.

Decimal: _____

SET 19 • 5.NBT.B.7

FORWARD 1

Add and subtract multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

Add.

$$128.47 + 9.605$$

Sum: _____

Subtract.

$$54.2 - 18.376$$

Difference: _____

Add.

$$0.952 + 3.48 + 11.6$$

A. 16.132

B. 4.53

C. 15.032

D. 16.032

Subtract.

$$100 - 0.485$$

Difference: _____

True or false?

$$7.3 - 2.145 = 5.245$$

A. True

B. False

SET 19 • 5.NBT.B.7

FORWARD 2

Add and subtract rational numbers • Grade 7 • The Number System •
7.NS.A.1.d

Find the value.

$$-12.5 + 4.75$$

Value: _____

Find the value.

$$3.2 - 8.45$$

Value: _____

Find the value.

$$-6.3 + (-2.85)$$

- A.** 9.15
- B.** -9.15
- C.** -3.45
- D.** 3.45

Find the value.

$$-4.6 - (-10.25)$$

Value: _____

True or false?

$$-1.5 + 1.5 = 0$$

A. True

B. False

Multiply decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiplying whole numbers	4.NBT.B.5
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
BACKWARD 3	Dividing by 10 and 100 to place the decimal point	5.NBT.A.2
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 20 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Multiply.

$$3.4 \times 6$$

Product: _____

Multiply.

$$0.7 \times 0.5$$

Product: _____

Multiply.

$$2.5 \times 1.2$$

A. 3

B. 30

C. 3.7

D. 0.3

Multiply.

$$0.06 \times 8$$

Product: _____

A pen costs \$1.25.

How much do 7 pens cost?

Cost: _____

SET 20 • 5.NBT.B.7

BACKWARD 1

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$34 \times 6$$

Product: _____

Multiply.

$$25 \times 12$$

Product: _____

True or false?

$$125 \times 7 = 785$$

A. True

B. False

$$18 \times 4 = ?$$

A. 72

B. 82

C. 48

D. 64

Multiply.

$$106 \times 8$$

Product: _____

SET 20 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 35 hundredths as a decimal.

Decimal: _____

True or false?

0.4 is 4 tenths.

A. True

B. False

Which decimal is 204 tenths?

- A.** 204
- B.** 20.4
- C.** 2.04
- D.** 0.204

How many hundredths are in 0.48?

Hundredths: _____

True or false?

0.06 is 6 tenths.

A. True

B. False

SET 20 • 5.NBT.B.7

BACKWARD 3

Dividing by 10 and 100 to place the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$$204 \div 10$$

Quotient: _____

Divide.

$$35 \div 100$$

Quotient: _____

True or false?

$$300 \div 100 = 30$$

A. True

B. False

$$875 \div 100 = ?$$

- A.** 875
- B.** 8.75
- C.** 0.875
- D.** 87.5

Divide.

$$48 \div 100$$

Quotient: _____

SET 20 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.45 \times 0.6$$

Product: _____

Multiply.

$$12.08 \times 2.5$$

Product: _____

Multiply.

$$0.125 \times 0.4$$

- A.** 0.5
- B.** 0.05
- C.** 5
- D.** 0.005

Multiply.

$$7.2 \times 1.15$$

Product: _____

True or false?

$$0.03 \times 0.03 = 0.09$$

A. True

B. False

SET 20 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-3.4 \times 6$$

Product: _____

Find the product.

$$(-0.7) \times (-0.5)$$

Product: _____

Find the product.

$$2.5 \times (-1.2)$$

A. -0.3

B. -30

C. 3

D. -3

Find the product.

$$(-0.06) \times 8$$

Product: _____

True or false?

$$(-1.25) \times (-4) = -5$$

A. True

B. False

Divide decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Dividing whole numbers	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 3	Multiplying both numbers by a power of 10 keeps the quotient	5.NBT.A.2
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 21 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Divide.

$$8.4 \div 4$$

Quotient: _____

Divide.

$$3.6 \div 0.6$$

Quotient: _____

Divide.

$$0.45 \div 0.05$$

A. 0.09

B. 0.9

C. 90

D. 9

Divide.

$$7.5 \div 0.3$$

Quotient: _____

A rope is 4.8 meters long. It is cut into pieces that are each 0.6 meter long.

How many pieces are there?

Pieces: _____

SET 21 • 5.NBT.B.7

BACKWARD 1

Dividing whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$84 \div 4$$

Quotient: _____

Divide.

$$360 \div 6$$

Quotient: _____

True or false?

$$450 \div 5 = 9$$

A. True

B. False

$$750 \div 3 = ?$$

- A.** 25
- B.** 250
- C.** 2,500
- D.** 260

Divide.

$$48 \div 6$$

Quotient: _____

SET 21 • 5.NBT.B.7

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $36 \div 6$ by thinking $6 \times \square = 36$.

Quotient: _____

True or false?

$7 \times 9 = 63$, so $63 \div 7 = 9$.

A. True

B. False

Which fact helps you find $56 \div 8$?

A. $56 \times 8 = 448$

B. $8 \times 7 = 56$

C. $8 + 48 = 56$

D. $8 - 7 = 1$

What number goes in the box?

$$\square \times 5 = 45$$

Number: _____

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

A. True

B. False

SET 21 • 5.NBT.B.7

BACKWARD 3

Multiplying both numbers by a power of 10 keeps the quotient • Grade 5 •
Number & Operations in Base Ten • 5.NBT.A.2

True or false?

$3.6 \div 0.6$ has the same quotient as $36 \div 6$.

A. True

B. False

Which has the same quotient as $0.45 \div 0.05$?

A. $4.5 \div 5$

B. $0.45 \div 5$

C. $45 \div 5$

D. $45 \div 0.5$

Multiply.

$$0.6 \times 10$$

Product: _____

Multiply.

$$0.05 \times 100$$

Product: _____

True or false?

$7.5 \div 0.3$ has the same quotient as $75 \div 30$.

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$13.65 \div 0.35$$

Quotient: _____

Divide.

$$9.072 \div 2.4$$

Quotient: _____

Divide.

$$0.081 \div 0.09$$

A. 90

B. 0.09

C. 0.9

D. 9

Divide.

$$52.5 \div 1.25$$

Quotient: _____

True or false?

$$1.8 \div 0.006 = 3$$

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-8.4 \div 4$$

Quotient: _____

Find the quotient.

$$(-3.6) \div (-0.6)$$

Quotient: _____

Find the quotient.

$$0.45 \div (-0.05)$$

- A.** -9
- B.** -0.9
- C.** 0.9
- D.** 9

Find the quotient.

$$-7.5 \div 0.3$$

Quotient: _____

True or false?

$$-(4.8 \div 0.6) = (-4.8) \div 0.6$$

A. True

B. False

Solve real-world problems with decimal operations

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Word problems with money and measurement	4.MD.A.2
BACKWARD 2	Hundredths on a grid	4.NF.C.6
FORWARD 1	Real-world problems with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 22 • 5.NBT.B.7

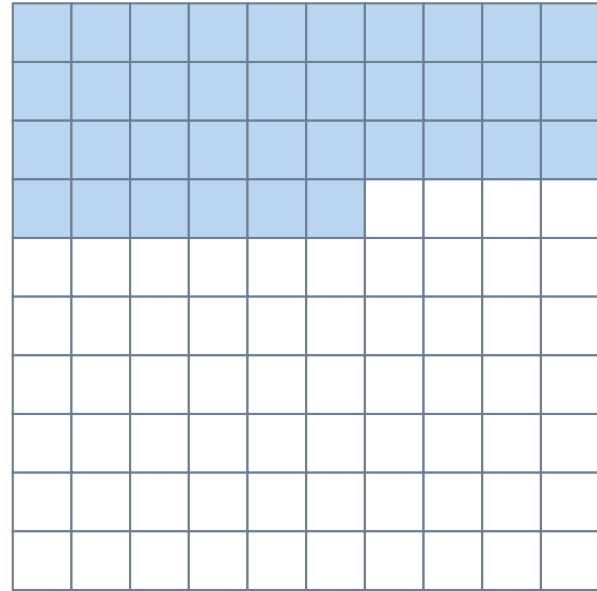
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.36 shaded.

How much more must be shaded to show 0.5?

Amount: _____



Juice costs \$2.35 a bottle. Ben buys 4 bottles and pays with a \$20 bill.

How much change does he get?

Change: _____

A 2.5-pound bag of nuts is shared equally among 5 friends.

How many pounds does each friend get?

- A.** 0.05 pound
- B.** 0.5 pound
- C.** 2 pounds
- D.** 12.5 pounds

A runner ran 3.75 km on Monday and 4.6 km on Tuesday.

How much farther did she run on Tuesday?

Distance: _____

Tiles cost \$0.45 each.

How much do 12 tiles cost?

Cost: _____

SET 22 • 5.NBT.B.7

BACKWARD 1

Word problems with money and measurement • Grade 4 • Measurement & Data • 4.MD.A.2

Pens cost 75 cents each.

How much do 4 pens cost, in dollars?

Cost: _____

A board is 120 cm long. It is cut into 4 equal pieces.
How long is each piece?

Length: _____

True or false?

Two quarters and three dimes are worth 65 cents.

A. True

B. False

Ann has \$5. She spends \$2.25.

How much money is left?

A. \$7.25

B. \$3.25

C. \$2.75

D. \$2.25

A jug holds 2 liters.

How many milliliters is that?

Milliliters: _____

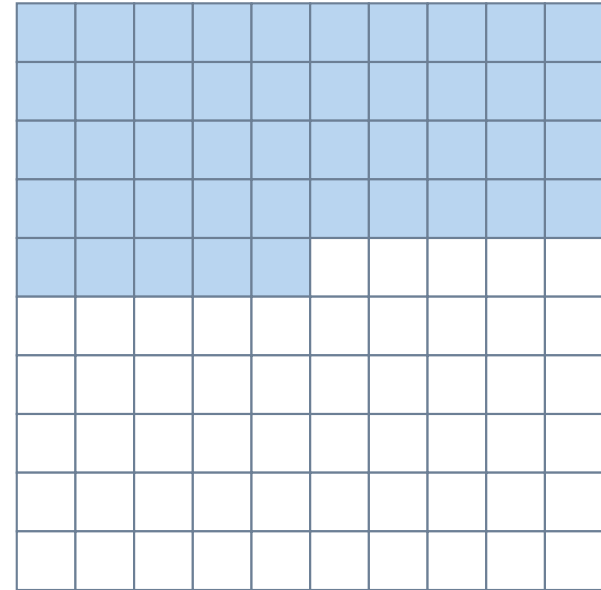
SET 22 • 5.NBT.B.7

BACKWARD 2

Hundredths on a grid • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal does the shaded part of the grid show?

Decimal: _____

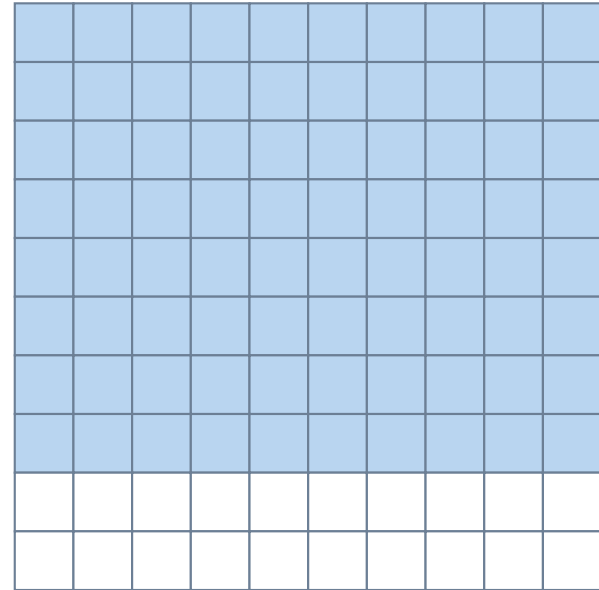


True or false?

The shaded part of the grid shows 0.8.

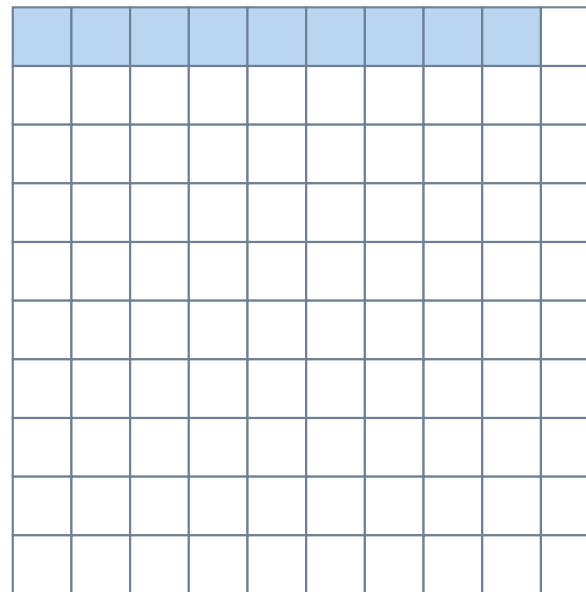
A. True

B. False



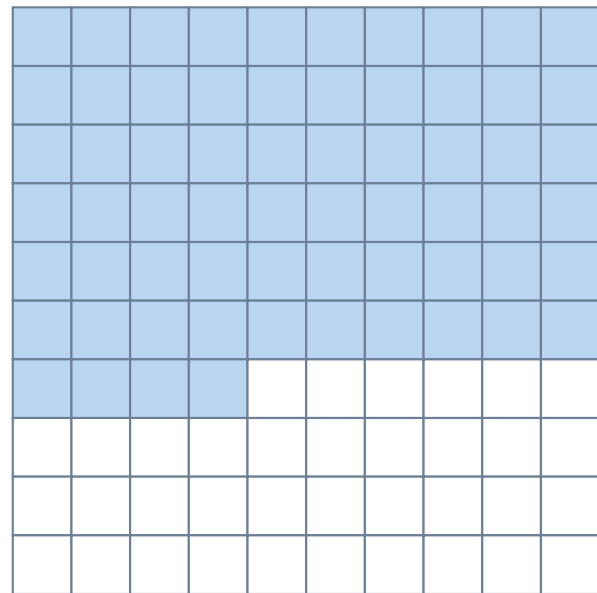
Which decimal does the shaded part of the grid show?

- A.** 0.9
- B.** 9
- C.** 0.009
- D.** 0.09



How many more squares
must be shaded to fill the
whole grid?

Squares: _____



True or false?

0.5 of a hundredths grid is 5 squares.

A. True

B. False

SET 22 • 5.NBT.B.7

FORWARD 1

Real-world problems with multi-digit decimals • Grade 6 • The Number System
• 6.NS.B.3

Gas costs \$3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

A board is 6.75 m long. It is cut into pieces that are 0.45 m long.

How many pieces are there?

Pieces: _____

Rice costs \$1.20 per pound.

How much do 2.5 pounds cost?

- A.** \$3.00
- B.** \$30.00
- C.** \$2.70
- D.** \$3.70

Lee buys items that cost \$12.49, \$3.05, and \$7.60.

What is the total cost?

Total: _____

True or false?

A 2.4-pound bag split into 0.3-pound portions makes 8 portions.

A. True

B. False

SET 22 • 5.NBT.B.7

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A submarine at -45.5 m rises 12.75 m and then dives 20 m.

What is its final position?

Position: _____

A store's profit was $-\$125.40$ in May and $\$310.15$ in June.

What was the total profit for the two months?

Total: _____

The temperature was 4°F . It dropped 1.5°F each hour for 6 hours.

What was the final temperature?

A. -5°F

B. 13°F

C. 5°F

D. -9°F

An account changes by $-\$2.40$ each week for 5 weeks.
What is the total change?

Total change: _____

True or false?

$$-3.6 \div 0.4 = -0.9$$

A. True

B. False

Add fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Adding fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Find the least common multiple	6.NS.B.4
FORWARD 2	Add rational numbers	7.NS.A.1.d

SET 23 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{2}{3} + \frac{1}{4}$$

Sum: _____

Add.

$$\frac{3}{5} + \frac{1}{10}$$

Sum: _____

Add.

$$\frac{5}{6} + \frac{3}{8}$$

A. $1\frac{5}{24}$

B. $1\frac{1}{24}$

C. $\frac{8}{24}$

D. $\frac{8}{14}$

Add.

$$\frac{1}{2} + \frac{2}{7}$$

Sum: _____

Add.

$$\frac{3}{4} + \frac{5}{12} + \frac{1}{6}$$

Sum: _____

SET 23 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{3} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{4} = \frac{4}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{3}{5}$?

A. $\frac{6}{5}$

B. $\frac{5}{3}$

C. $\frac{3}{10}$

D. $\frac{6}{10}$

Find the missing number.

$$\frac{5}{6} = \frac{20}{?}$$

Missing number: _____

True or false?

$$\frac{2}{7} = \frac{4}{21}$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 2

Adding fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{3}{8} + \frac{2}{8}$$

Sum: _____

True or false?

$$\frac{4}{12} + \frac{5}{12} = \frac{9}{24}$$

A. True

B. False

$$\frac{7}{10} + \frac{6}{10} = ?$$

A. $1\frac{6}{10}$

B. $1\frac{3}{10}$

C. $\frac{13}{20}$

D. $\frac{1}{10}$

Add.

$$\frac{1}{6} + \frac{4}{6}$$

Sum: _____

True or false?

$$\frac{2}{5} + \frac{3}{5} = 1$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 4.

Multiples: _____

True or false?

10 is a multiple of both 3 and 4.

A. True

B. False

Which number is a multiple of both 6 and 8?

A. 16

B. 30

C. 24

D. 14